

Deep learning and its applications in nuclear magnetic resonance spectroscopy

Yao Luo, Xiaoxu Zheng, Mengjie Qiu, Yaoping Gou, Zhengxian Yang, Xiaobo Qu, Zhong Chen, Yanqin Lin^{*}

Fujian Provincial Key Laboratory of Plasma and Magnetic Resonance, Department of Electronic Science, State Key Laboratory of Physical Chemistry of Solid Surfaces, Xiamen University, Xiamen 361005, China

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ABSTRACT

Nuclear Magnetic Resonance (NMR), as an advanced technology, has widespread applications in various fields like chemistry, biology, and medicine. However, issues such as long acquisition times for multidimensional spectra and low sensitivity limit the broader application of NMR. Traditional algorithms aim to address these issues but have limitations in speed and accuracy. Deep Learning (DL), a branch of Artificial Intelligence (AI) technology, has shown remarkable success in many fields including NMR. This paper presents an overview of the basics of DL and current applications of DL in NMR, highlights existing challenges, and suggests potential directions for improvement.

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Abbreviations: AF2, AlphaFold2; AI, artificial intelligence; ANNs, artificial neural networks; BMRB, biological magnetic resonance data bank; BP, back-propagation; CEST, chemical exchange saturation transfer; CNN, convolutional neural network; CReSS, cross-modal retrieval between spectrum and structure; CrR, creatine referencing; CSD, cambridge structural database; dCrR, deep learning-based creatine referencing; DFT, density functional theory; DL, deep learning; dSR, deep learning-based spectral registration; FCNNs, fully connected neural networks; FID, free induction decay; FPC, frequency-and-phase correction; GABA, gamma-aminobutyric acid; GNN, graph neural network; GPU, graphics processing unit; IST, iterative soft thresholding; KRR, kernel ridge regression; LF-NMR, low-field nuclear magnetic resonance; LSTM, long short-term memory; ML, machine learning; MLP, multi-layer perceptron; MRS, magnetic resonance spectroscopy; NLP, natural language processing; NMR, nuclear magnetic resonance; NSA, number of signal average; NUS, non-uniform sampling; PCA, principal component analysis; PCC, pearson correlation coefficient; PDB, protein data bank; PLS, partial least-squares; PRESS, point resolved spectroscopy; QM, quantum mechanical; RMSD, root mean square deviation; RNN, recurrent neural network; SCC, scalar coupling constant; SNR, signal-to-noise ratio; SVM, support vector machines.

^{*} Corresponding author.

E-mail address: linyq@xmu.edu.cn (Y. Lin).

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1. Introduction

Nuclear Magnetic Resonance (NMR) is a powerful analytical technique that is widely used to obtain valuable information on molecular composition, structure, and dynamics. It has been applied in multiple disciplines of research, such as chemistry, biology, medicine, materials science, etc. However, several existing challenging issues limit its broader applications. For example, the sampling time of multidimensional NMR spectroscopy grows exponentially with the number of indirect dimensions. Long experiment times prevent NMR applications in some circumstances, such as unstable samples or limited spectrometer time availability. NMR spectra often suffer from intrinsically low Signal-to-Noise Ratio (SNR), especially for nuclei with low natural abundance, such as ^{13}C . Low spectral SNR is typically associated with high levels of noise, which obscures weak signals and makes spectral analysis more challenging. In addition, spectra of large molecular systems or complex mixtures containing many peaks that tend to overlap with each other make spectral deconvolution challenging without expert assistance.

Many signal-processing methods have been proposed to solve these issues. For example, Non-Uniform Sampling (NUS) technology is a commonly used approach to reduce time cost, and various algorithms have been developed to reconstruct NUS data, such as Iterative Soft Thresholding (IST) [1], Sparse Multidimensional Iterative Lineshape-Enhanced (SMILE) [2], CLEAN [3], Maximum Entropy [4], Multi-Dimensional Decomposition (MDD) [5] and Covariance [6]. A resampling method (NASR) [7] has been proposed to suppress noise. These conventional algorithms can solve the relevant problems in the NMR field to a certain extent, but shortcomings such as reliance on prior knowledge accompany them and long processing times. Over recent years, Artificial Intelligence (AI) has developed rapidly, with remarkable achievements being noted in multiple fields. In 2024, the Nobel Prize in Physics was awarded to American scientist John J. Hopfield and British-Canadian scientist Geoffrey E. Hinton for their outstanding contributions to the field of AI. As a branch of AI, the rapidly-evolving Deep Learning (DL) technique also achieved great success in many fields, offers a promising way to solve problems in NMR. Particularly, DL achieved significant breakthroughs in the prediction of protein [8,9] and RNA [10] structures, as represented by AlphaFold [11], developed by DeepMind. Its release has had a significant impact on the field of protein structure prediction, providing a crucial tool for biomedical research

and drug development. In 2024, the Nobel Prize in Chemistry was awarded to David Baker, John M. Jumper and Demis Hassabis, in recognition of their contributions to the protein design and structure inference based on AI. In general, the powerful capabilities of DL have driven progress in a variety of fields including NMR. This paper reviews the basics of DL and the utilization of DL for processing and analyzing NMR data.

2. History of deep learning

DL refers to the training of multi-layer Artificial Neural Networks (ANNs). It is a subset of Machine Learning (ML) and a crucial approach for achieving AI. The roots of DL can be traced back to the 1950 s, when the binary single neuron model [12] was created by Rosenblatt. This model was inspired by the biological principle of a neuron in organisms, and became the precursor to contemporary neural networks. The basic element of a neural network is the *neuron*. Each neuron receives as input either the input to the network, or outputs from other neurons in previous layers; applies a transformation by some non-linear activation function (a nonlinear transformation applied by each neuron in the neural network to determine its output based on the input it receives); and passes the transformation result either to the neurons in the subsequent layer (a *layer* is a collection of neurons) or to the network output. Neural networks are typically stacked by layers, which can be categorized as input layer, hidden layer and output layer. However, the limitations of a basic neural network, a “single perceptron model”, in solving non-linear problems (non-linear problem means that the relationship between inputs and outputs of the network cannot be accurately modeled as a linear function) caused a long period of stagnation in neural network research. In 1986, Rumelhart et al. [13] proposed the BackPropagation (BP) algorithm, suitable for Multi-Layer Perceptrons (MLPs). MLPs were the earliest Fully Connected Neural Networks (FCNNs), and serve as the foundational structure for DL. In addition, Rumelhart et al. used a sigmoidactivation function (a mathematical function that maps any real-valued number into a value between 0 and 1, which introduces non-linearity) to train their network. This approach allows the network to learn complex (non-linear) relationships between inputs and outputs of the network, and sparked a resurgence in neural network research. In 1989, LeCun et al. [14] proposed a FCNN to address practical problems in handwritten digit (MNIST) recognition

tasks. Subsequently, LeCun et al. proposed LeNet-5 [15], a Convolutional Neural Network (CNN) that introduced the convolutional layer, down-sampling layer, and fully connected layer, and defined the basic structure of modern CNNs. The convolutional layer is fundamental in a CNN, extracting features through convolution operations. The down-sampling layer reduces the spatial dimensions (width and height) of the feature maps while retaining the most important information, to reduce computational load. The fully connected layer, typically located at the network's end, combines features learned in the previous layers to perform the final task. The emergence of CNNs was of great importance. They efficiently capture local features within images, and significantly reduce the number of parameters that need to be learned, through weight sharing (weights are values associated with the connections between neurons, which determine the strength and sign of the influence one neuron has on another).

While CNNs effectively capture local features using convolutional kernels, they are not suitable for all types of data. For sequence data and time-related tasks, CNNs may ignore global context information, leading to imprecise predictions. The emergence of the Recurrent Neural Network (RNN) solved this problem. In 1990, Elman [16] proposed the simplest form of a RNN, and trained it using the BP algorithm. The RNN has unique advantages in handling sequential data, time-dependent tasks, and certain multimodal applications. It should be noted that the BP algorithm suffers from the problem of *vanishing gradients*. Here *gradient* is the direction and rate of change of the loss function, which describes the difference between the output data and the label data, with respect to weights. The vanishing gradient problem is where gradients become extremely small, leading to slow convergence or even stagnation, hindering the training process. The converse problem is that of *exploding gradients*, where gradients grow uncontrollably, resulting in numerical instability and divergence, and again destabilizing training. Unfortunately, limited computational power and data availability led to a decline in neural network research in favor of other statistical methods such as Support Vector Machines (SVM) [17], decision tree models [18], and boosting [19]. The breakthrough in Graphics Processing Unit (GPU) technology has since met the computational needs of neural networks. This advance in computing power has lifted the limitations on neural networks. In 2006, Hinton et al. [20] developed the Deep Belief Network (DBN) and introduced the concept of DL, which brought neural networks back into focus.

With the continuous advances in hardware technology and the emergence of large-scale datasets, DL has undergone rapid development and widespread application in various fields, particularly in Computer Vision (CV) for tasks such as object detection [21], image segmentation [22], and facial recognition [23]. In the 2012 ImageNet competition, a CNN network called AlexNet [24] defeated all other competitors, bringing DL technology into the spotlight. The competition required the classification of a dataset consisting of over one million images, and AlexNet achieved extraordinary results in this task. As for the field of Natural Language Processing (NLP), DL have demonstrated significant

success in a range of tasks, such as text classification [25], sentiment analysis [26], machine translation [27], and speech recognition [28]. The DL-based Word2Vec technique launched by Google in 2013 brought a fundamental breakthrough in the NLP field, providing semantically rich word representations. Recently, the emergence of ChatGPT [29], a dialogue generation model developed by OpenAI, has become a milestone in NLP. It is based on the Generative Pre-trained Transformer (GPT)-3 architecture and focuses on text generation through dialogue. ChatGPT has the ability to generate natural conversations that can be applied to a variety of topics and scenarios.

3. Basics of deep learning

Before utilizing a neural network for processing real-world data, it is crucial to train the network to converge, using large datasets. Fig. 1 shows a common training process for a neural network. The fundamental process for network convergence utilizes forward propagation and backpropagation algorithms. During forward propagation in a neural network, the input layer receives the input data and passes the processed result (output) to the hidden layers. Each hidden layer subsequently processes the output of the previous layer until the data are propagated to the output layer. Subsequently, the loss function is used to calculate the difference between the output data and the label data, and this loss is backpropagated through the network, starting from the output layer to compute the gradients [13]. Based on these gradients, optimization algorithms such as Stochastic Gradient Descent (SGD) [30], Adam [31], AdaGrad [32], and Momentum [33] update all the weights from initial values. Additionally, to further improve the performance of the neural network model, regularization techniques (such as L1 regularization [34], L2 regularization [35], dropout [36], etc.), activation functions (such as sigmoid [13], ReLU [37], etc.), and Batch Normalization (BN) [38] are typically employed. After multiple epochs (an *epoch* is one complete pass through the entire training dataset during the training process, starting from forward propagation and ending with backward propagation) of training, a well-trained model has a small value of loss and achieves convergence, and it can effectively accomplish the corresponding tasks.

DL models utilize different network architectures depending on the task. In applications in the NMR field, DL mainly employs three basic architectures: FCNN, CNN, and RNN. FCNNs are good at establishing inherent correlations in the whole data, and thus were often used for classification and quantitative analysis in NMR before CNNs and RNNs were introduced into the field of NMR; CNNs do well in delineating spatial information in data, especially local information, and thus can be utilized for denoising, reconstruction, inference, peak picking and so on; and RNNs are adept at sequential data analysis, so can be used to process sequential NMR data, especially Free Induction Decays (FIDs). Choosing and adapting a suitable network architecture according to the characteristics of the data and task is the first and most important step when applying DL to NMR. The next section will provide a detailed

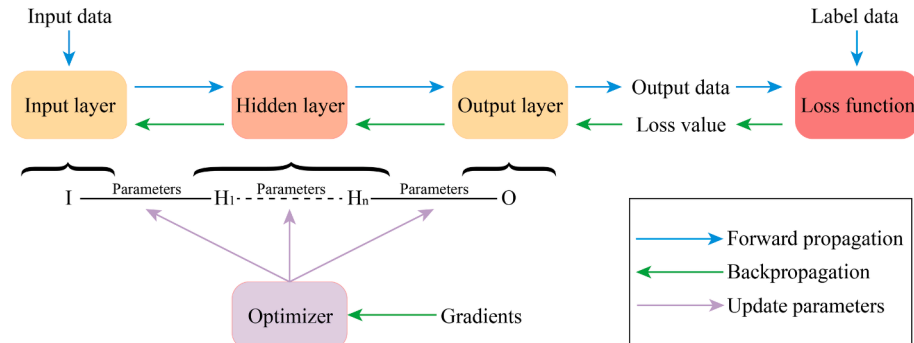


Fig. 1. The training process of a common neural network model. I , H_i , and O represent the input layer, the i -th hidden layer, and the output layer, respectively.

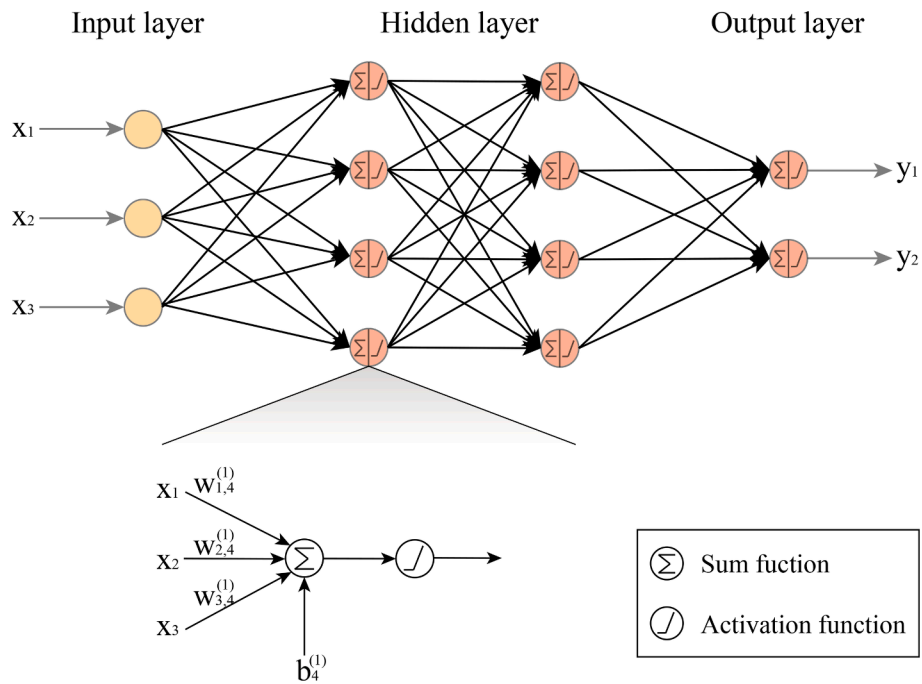


Fig. 2. A simple FCNN with two hidden layers. The fourth neuron of the first hidden layer is expanded. $W_{i,j}^{(k)}$ represents the weight between the i -th neuron in the $(k-1)$ -th layer and the j -th neuron in the k -th layer. $b_j^{(k)}$ represents the bias (a value that is added to the weighted sum of inputs before an activation function is applied.) of the j -th neuron in the k -th layer.

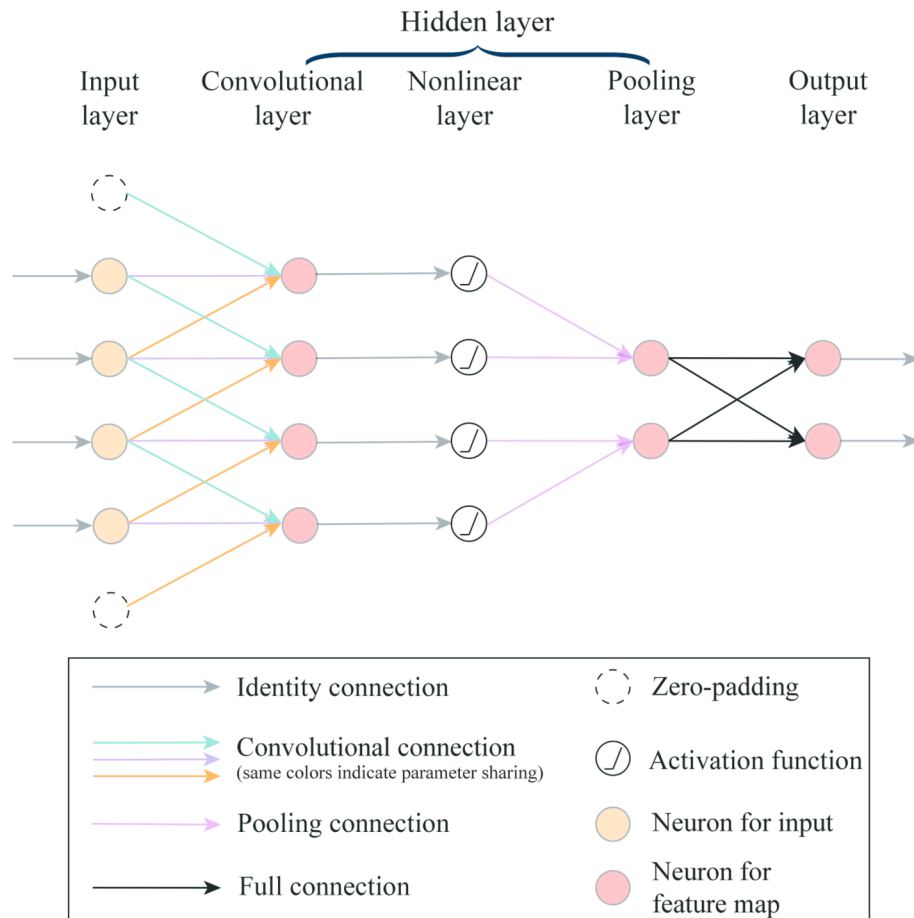


Fig. 3. A simple CNN structure. Identity connection allows the direct flow of features or information between layers without any process. Convolutional connections with the same color share the same parameters (weights and biases). Zero padding makes the size of the output feature map of the convolutional layer consistent with the input.

introduction to these three basic DL architectures.

3.1. Fully connected neural networks (FCNNs)

The FCNN is a fundamental artificial neural network structure in which each neuron connects to all neurons in the previous and subsequent layers. The data propagates unidirectionally from the input layer to the output layer, without convolution operations or temporal connections. For example, in a simple FCNN with two hidden layers, as shown in Fig. 2, every neuron in a given layer is connected to all neurons in the previous and following layer. Each neuron in an FCNN takes in inputs and then inputs are weighted, summed, biased, and then non-linearized through an activation function. This allows the network to effectively handle non-linear problems. It is important to note that the input layer does not contain processing operations and is therefore typically not included when counting layers. For example, the FCNN in Fig. 2 with two hidden layers and an output layer is referred to as a 3-layer FCNN. When an FCNN has many hidden layers, it runs a risk of overfitting. To mitigate this issue, dropout layers are often employed following the fully connected layers. These dropout layers randomly drop some of the neurons in the layers during each training iteration, preventing the network from relying too heavily on any individual neuron.

As the earliest DL network, FCNN possesses certain advantages. Specifically, with employing non-linear activation functions and multiple layers, FCNN exhibits strong non-linear fitting capabilities compared to previous non-DL methods, enabling it to analyze complicated NMR data. Furthermore, the fully connected layers within FCNNs facilitate the extraction of global features. Despite these strengths, FCNNs suffer from several limitations. They struggle to learn overly complex or abstract representations, have an excessive number of parameters, and rely greatly on skilled training. These challenges can be mitigated through the use of CNNs.

3.2. Convolutional neural networks (CNNs)

CNNs are the most common type of neural networks, and convolutional layer, non-linear layer, and pooling layer are three common hidden layers in CNNs. Among these hidden layers, convolutional layer is the most essential, extracting features from input data through convolution operations and providing valuable information for

subsequent layers. In some modern algorithms, more complex blocks like the inception block [39–41] and residual block [42] are employed.

The structure of a simple CNN is illustrated in Fig. 3. In contrast to neurons in FCNNs, each neuron in the convolutional layers of CNNs is only connected to a portion of neurons of the previous layer, referred to as a receptive field. Convolutional layers extract features by applying convolution operations using filters. A filter is capable of extracting features for all local regions of a feature map, thereby enabling parameter sharing. Similar to FCNNs, the feature maps generated by the convolutional layers are processed through a non-linear activation function. Pooling layer is used for down-sampling and generally involves max pooling and average pooling. Pooling helps reduce the size of feature maps and the number of parameters, which in turn accelerates computation and prevents overfitting. The fully connected layer is typically employed as the output layer to obtain the weighted sum of the feature maps. However, due to the redundant parameters of fully connected layers, modern network architectures such as ResNet [42] and GoogLeNet [39] utilize a global average pooling layer as the output layer instead.

Additionally, to expedite network convergence, avoid vanishing or exploding gradients, and prevent overfitting, a BN layer is frequently employed following convolutional layers. Compared to FCNNs or previous networks, convolutional layers in CNNs utilize filters to process locally connected regions with identical parameters, resulting in fewer parameters and higher computational efficiency. Moreover, the CNN has translational and rotational invariance, making it suitable for analyzing more complex NMR data, like inferring chemical shifts from input NMR spectra. Compared to FCNNs, CNNs are adept at analyzing local information and detecting signals benefiting from the local feature extraction provided by convolution kernels, which may be beneficial for identifying small NMR spectral peaks.

3.3. Recurrent neural networks (RNNs)

RNNs are good at processing sequential input data. In this type of data, subsequent data points are correlated with previous ones; and such correlation cannot be efficiently extracted using convolution in CNNs or weighted sum in FCNNs. RNNs overcome this limitation by incorporating recurrent units, allowing for efficient processing of sequential data. Therefore, RNNs are frequently used in the time domain, particularly in NLP. Fig. 4 illustrates a simple RNN containing a single hidden

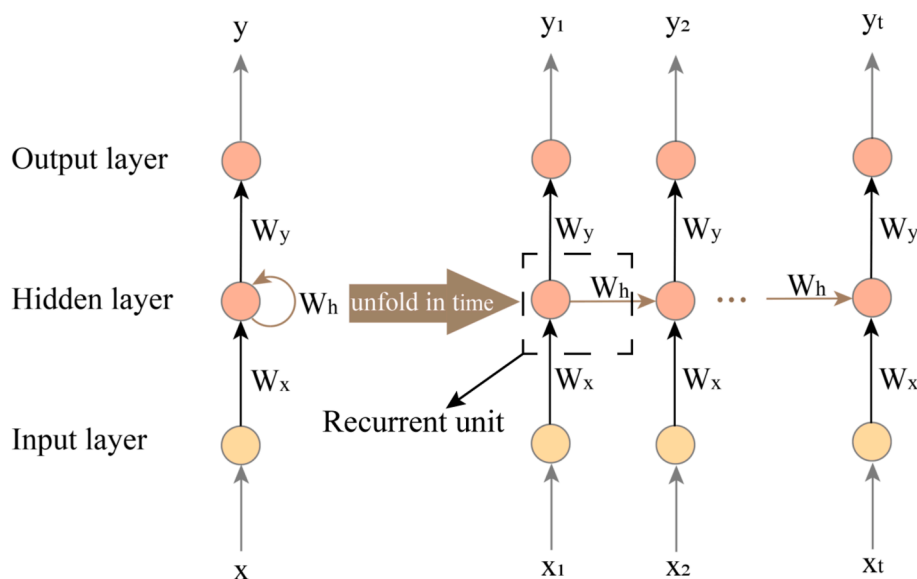


Fig. 4. A simple RNN with one hidden layer. An RNN can be unfolded in time. x_i and y_i are the input and the output at the i -th time point, respectively. W_x , W_y , and W_h are the weights between different neurons. The dashed rectangle represents a recurrent unit.

layer. RNNs are capable of processing one element of sequential data at a time. At each time step, a recurrent unit in the hidden layer takes as input not only the output of the previous layer, but also its output at the previous time step. This allows RNNs to extract contextual information from earlier parts of the sequence, enabling them to capture dependencies over time effectively. The parameters (W_x , W_y , and W_h) remain constant throughout forward propagation, and are updated through BackPropagation Through Time (BPTT) [43]. Here W_x is the weight from neurons in the input layer to neurons in the first hidden layer, W_h is the weight from neurons in the current hidden layer to neurons in the next hidden layer, and W_y is the weight from neurons in the hidden layer to neurons in the output layer.

Unlike feed-forward neural networks such as FCNNs and CNNs, RNNs possess short-term memory, due to the use of recurrent units. However, traditional RNNs suffer from the problems of vanishing and exploding gradients, rendering them incapable of processing long time sequences [44]. A variation of RNN, known as the Long Short-Term Memory (LSTM) network [45], overcomes these problems by incorporating a gating mechanism that integrates short-term memory with long-term memory. There are several other enhanced versions of RNNs: Recurrent Convolutional Neural Network (RCNN) [46], which combines the strengths of RNNs and CNNs, Recursive Neural Network (RecNN) [47], which extends RNNs into tree structures, and Graph Neural Network (GNN) [48], which extends RNNs into graph structures.

Recently, attention mechanisms [49] and transformers [50], widely used in the field of NLP, have been combined with basic networks, bringing a revolution to DL. It is envisioned that these algorithms will further improve NMR spectral quality.

4. Deep learning applications in traditional NMR

4.1. Reconstruction of non-uniform sampled spectroscopy

4.1.1. Multidimensional spectroscopy

Reconstruction algorithms for non-uniform sampling multidimensional spectroscopy can be broadly categorized into traditional and DL-based approaches. Common traditional algorithms include IST [1], SMILE [2], CLEAN [3], Maximum Entropy [4], MDD [5], and Covariance [6]. Currently, the IST and SMILE algorithms are those most commonly employed. Traditional algorithms demonstrate effectiveness in reconstructing non-uniform sampling data, but they still have certain limitations: 1, the application of traditional algorithms necessitates prior knowledge or theoretical assumptions; 2, traditional algorithms frequently demand multiple iterative optimizations, leading to high computational complexity and considerable time cost, especially when processing large-scale data; 3, several key parameters in traditional algorithms significantly impact the results, and it is hard to determine optimal parameter values in practical applications, which may result in suboptimal reconstruction outcomes. The emergence of DL has

presented new opportunities to overcome these limitations.

DL relies on having a sufficiently large amount of data to support the entire training process. Using too limited a quantity of training data could yield a trained DL model that lacks adequate generality. However, obtaining a sufficient volume of real experimental NMR data are time-consuming, and often simulated data can be employed instead. It is known that Free Induction Decay (FID) signals can be characterized as exponential decay signals. For instance, a one-dimensional FID signal can be mathematically represented by the following equation:

$$FID(t) = A \exp(i2\pi f t) \exp(-t/T_2). \quad (1)$$

Similarly, the two-dimensional FID signal can be expressed as

$$FID(t_1, t_2) = A \exp(i2\pi f_1 t_1) \exp(-t_1/T_2^{(1)}) \times \exp(i2\pi f_2 t_2) \exp(-t_2/T_2^{(2)}). \quad (2)$$

Here A , f , and T_2 are the amplitude, frequency, and transverse relaxation time. In addition, $t_i = 1/SW_i \times (0:1:N-1)$, where SW and N are bandwidth and the number of complex data points respectively. With this theoretical basis, significant volumes of simulated NMR data can be generated, as demonstrated by Qu et al. [51], effectively mitigating the challenge of the limited quantity of NMR training data.

Presently, DL based NUS reconstructions were performed on single-domain, specifically in the time or frequency domain, as shown in Fig. 5. In time-domain reconstruction, simple convolutional techniques are unable to effectively extract features of the temporal correlation present within the signal. Consequently, DL networks such as LSTM [45], RNN [52], and WaveNet [53], which are frequently employed for time-domain signals like speech, are utilized to reconstruct NUS under-sampled signals. Hansen [54] proposed the utilization of DL networks based on the LSTM architecture for the reconstruction of time-domain NMR signals, using both undersampling schemes and undersampled data as inputs. In a follow-up study, Karunanithy et al. [55] presented an FID-Net network based on the WaveNet architecture, as shown in Fig. 6. WaveNet employs dilated convolutions, a technique that expands the receptive field to capture more context in feature extraction. This technique better connects the causal relationships in time-series data, leading to improved reconstruction quality of time-domain signals. Intuitively, reconstruction in the time domain allows direct handling of undersampled FID signals, offering specific benefits in accurately reconstructing peak intensity. However, as the proportion of points sampled decreases, the amount of information from the undersampled time-domain signal declines, resulting in a rapid decline in the reconstruction performance of DL methods.

In addition to time-domain reconstruction, frequency-domain reconstruction is also a hot research topic. When reconstructing frequency-domain data, a significant number of artifacts are generated in the Fourier transform of the raw data after the replacement of missing time-domain points with zeros. The role of DL-based frequency-domain

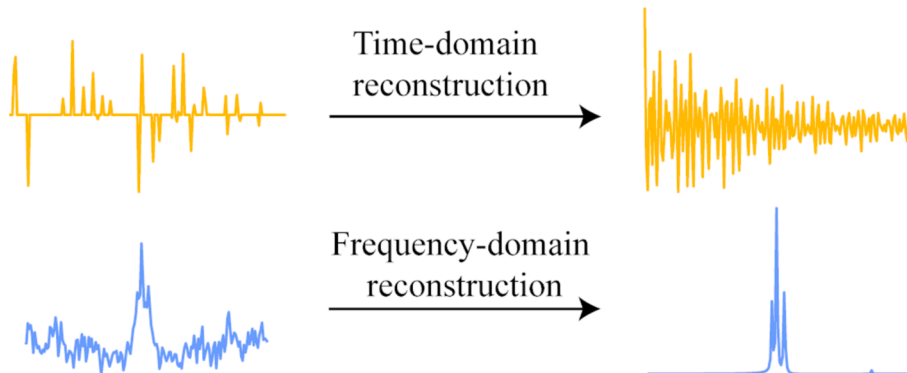


Fig. 5. NMR NUS reconstruction in time and frequency domains by DL.

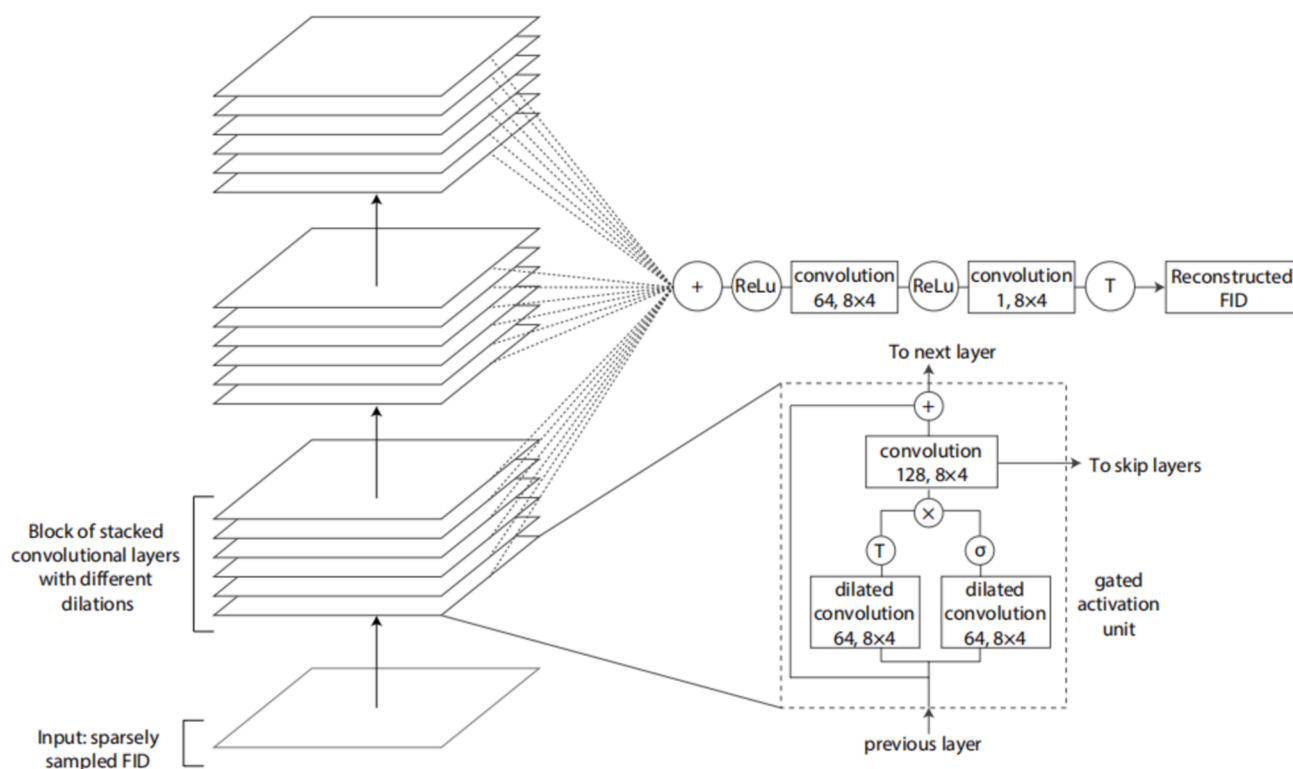


Fig. 6. The architecture of FID-Net. Relu refers to the rectified linear unit, T refers to hyperbolic tangent and sigmoidal activation functions, and σ refers to sigmoidal activation functions.

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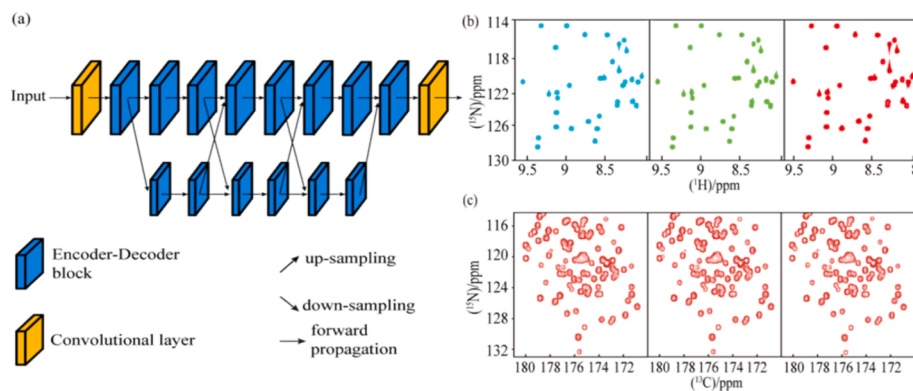


Fig. 7. (a) Network structure of EDHRN. (b) Reconstructed HSQC spectrum of GB1 protein with the proportion of points sampled is 25 %: from left to right are the fully sampled reference spectrum, EDHRN reconstructed spectrum, and hmsIST reconstructed spectrum. (c) Projection of reconstructed HNCO 3D reconstructed spectrum of Azurin protein with the proportion of points sampled is 8 %: from left to right are the fully sampled reference spectrum, EDHRN reconstructed spectrum, and hmsIST reconstructed spectrum.

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reconstruction is to remove these undersampling artifacts. In the realm of DL-based frequency-domain methods, Qu et al. used DenseNet [51], Deep Hankel Matrix Factorization (DHMF) network [56], and a deep thresholding network [57] to reconstruct frequency-domain signals in each case. Luo et al. [58] proposed the EDHRN network, based on HRNet [59], and integrated a virtual echo module for data preprocessing to accomplish frequency-domain reconstruction, as shown in Fig. 7. Similarly to FID-Net [55], based on WaveNet architecture, Jahangiri et al. [60] treated the NUS reconstruction task as a pattern recognition problem for feature extraction in the frequency domain. The objective of frequency-domain reconstruction is to eliminate undersampling artifacts from the spectrum while retaining the real peaks. In general the

undersampling artifacts derived from a given peak in the frequency domain are weaker than the peak itself, and thus it is easy to distinguish the real peak from its corresponding artifacts in a single peak spectrum. However, a spectrum typically consists of multiple peaks with different amplitudes, and the artifacts caused by strong peaks may drown out the weak peaks. This can bring great difficulties in the identification of weak peaks, and can even make it impossible to distinguish weak peaks from artifacts, leading to peak loss or artifacts in the reconstruction results. Time-domain and frequency-domain reconstructions have different advantages and disadvantages. Therefore, it seems feasible to develop a DL model that leverages the strengths of both time- and frequency-domain reconstructions while overcoming their individual limitations, as a

Table 1

Technical details of different DL reconstruction algorithms.

Author	Network Foundation	Domain*	Evaluation Index	Sampling Scheme
Present author's lab. [58]	HRN [59]	Frequency	PCC	Poisson-Gap
Hansen [54]	LSTM [45]	Time	PCC	Poisson-Gap Random-Gap
Karunanithy et al. [55]	WaveNet [53]	Time	PCC RMSD	Poisson-Gap Random-Gap
Wang et al. [57]	DRSNs [65]	Frequency	PCC RMSD	Poisson-Gap
Huang et al. [56]	DenseNet [66]	Frequency	PCC	Poisson-Gap
Jahangiri et al. [60]	WaveNet [53]	Frequency	PCC RMSD	Poisson-Gap
Qu et al. [51]	DenseNet [66]	Frequency	PCC	Poisson-Gap

* The column shows in which domain the reconstructions are performed. Pearson Correlation Coefficient (PCC), Root Mean Square Deviation (RMSD).

promising research direction.

While the algorithm's strength is undoubtedly crucial, the choice of the undersampling scheme can also have a significant impact on the quality of the reconstructed spectrum. Exponential random sampling was first proposed by Barna et al. [61] in 1987, and produces a sampling pattern that is dense in the beginning and sparse later on. This approach is based on the fact that the FID signal is strong in the early stages of evolution and gradually decays in the later stages, resulting in better reconstruction performance compared to random sampling. However, this method may cause large gaps between sampling points, leading to increased reconstruction artifacts [62]. Poisson-Gap sampling solves this problem to some extent, as proposed by Hyberts et al. [4] in 2010. In [4], a Poisson-distributed interval is employed to avoid large gaps between

sampling points. Currently, these sampling schemes are still widely used in verifying the performance of DL reconstruction algorithms. As shown in Table 1, Qu et al. [51], Huang et al. [56], Wang et al. [57], Hansen and Karunanithy [54,55], Luo et al. [58], and Jahangiri et al. [60] conducted reconstruction experiments using the Poisson-Gap sampling scheme. In addition, Karunanithy et al. [55], and Jahangiri et al. [60] utilized the random sampling scheme for reconstruction and compared it with the Poisson-Gap sampling scheme. The experimental results demonstrate that Poisson-Gap sampling has a significant advantage.

Current methods for evaluating reconstructed spectrum quality entail computing the Pearson Correlation Coefficient (PCC) or the Root Mean Square Deviation (RMSD) between the reconstructed and the fully sampled spectrum or evaluating the incidence of missing peaks or artifacts by comparing the reconstructed spectrum to the fully sampled spectrum. However, these evaluation methods require the availability of a fully sampled spectrum for comparison, which is often difficult or even impossible to obtain in practice. Therefore, the assessment of reconstructed results remains a challenge, particularly when fully sampled spectra are not available. Fortunately, the method of DL-based no-reference (not requiring a reference image) image quality assessment [63,64] may provide a way to assess the quality of reconstructed NMR spectra.

There are still many challenges that need to be overcome in the reconstruction of NMR spectra. In terms of reconstruction accuracy, the amplitude difference between the highest and lowest peaks can exceed one hundredfold in high dynamic range spectra, making it difficult for DL reconstruction algorithms to capture small spectral peaks. In terms of the universality of reconstruction algorithms, designing DL networks and reconstruction strategies that are capable of reconstructing spectra with varying proportions of points sampled, schemes, and sizes remains a significant challenge. In terms of higher dimensional spectra (such as 4D and 5D spectra), there is currently no mature DL method available

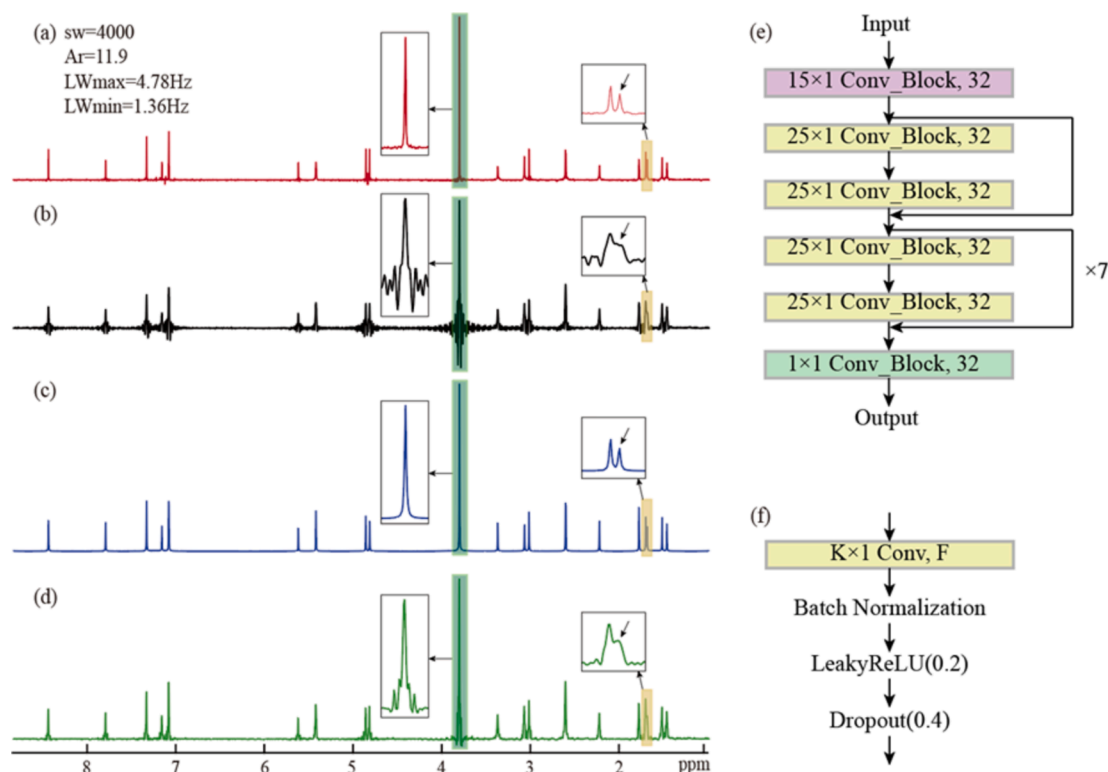


Fig. 8. Spectra of quinine with a concentration of 0.05 M. (a) Fully sampled PSYCHE pure shift spectrum (1 min and 35 s), (b) PSYCHE pure shift spectrum with the proportion of points sampled is 15.6 % (15 s), (c) spectrum reconstructed using PS-ResNet, (d) spectra reconstructed using the traditional algorithm IST. (e) is the overall network architecture of PS-ResNet, and (f) is the structure of Conv_Block. Reproduced with permission from [68].

for the task of reconstruction.

4.1.2. 1D pure shift spectroscopy

The NUS undersampling reconstruction technique can be applied not only to real two-dimensional spectra, but also to pseudo-two-dimensional spectra obtained through pseudo-two-dimensional sampling. For instance, in the acquisition of pure chemical shift spectra, PSYCHE [67] employs a pseudo-two-dimensional acquisition mode that requires a long acquisition time. To efficiently obtain high-quality pure chemical shift spectra, Zheng et al. [68] used NUS to reduce the number of points in the experiment and then reconstructed the undersampled spectra using a PS-ResNet network based on the ResNet architecture. The network framework of PS-ResNet and the experimental results from a quinine sample are shown in Fig. 8. The proportion of points sampled is as low as 15.6 % and the acquisition time of the PSYCHE pure shift spectrum of quinine is reduced from 1 min and 35 s to 15 s. Compared to traditional methods like IST, the spectrum reconstructed by PS-ResNet effectively removes undersampling artifacts and restores the peak shape, resulting in a cleaner spectrum. This demonstrates the excellent performance of DL in reconstructing undersampled spectra at the low proportion of points sampled. Differing from Zheng et al., Shen et al. [69] implemented rapid sampling for both PSYCHE [67] and original ZS [70] (Zangger-Sterk, also known as slice-selective) spectra using NUS. They developed a UNet-based T2F (time-to-frequency) network for reconstructing NUS spectra, achieving good performance at a 37.5 % proportion of points sampled.

PS-ResNet, proposed by Zheng et al., enables the rapid acquisition of pure chemical shift spectra. However, their DL method relies on a substantial amount of training data and extended training time, which is a common issue with current DL methods. With the emergence of lightweight neural networks, DL methods can be quickly trained with minimal training data, overcoming the usual constraints of requiring extensive training data and high-performance computing resources (such as GPUs). Building upon lightweight neural networks, Zhan et al. introduced a lightweight [71] and a lightweight attention-assisted [72] DL method, trained on a dataset of several hundred simulated data, achieving outstanding performance on personal computers.

4.2. Denoising

NMR is less sensitive than other spectroscopic techniques such as mass spectroscopy. It suffers from low Signal-to-Noise Ratio (SNR), especially for nuclei with low natural abundance such as ^{13}C . Suppressing noise is an effective way to increase SNR. Kobayashi et al. [73] demonstrated that a CNN that consists of two convolutional layers can suppress noise, but careful design of the network structure is needed to suppress stronger noise. Wu et al. [74] proposed DeNoising Unet (DN-Unet) to suppress strong noise. The network used the encoder-decoder structure and developed an M-to-S strategy in which multiple noise spectra (inputs) correspond to the same noise-free spectrum (label) during the training stage. High-quality denoising of 1D, 2D, and 3D spectra can be achieved by DN-Unet, using the same trained 1D model,

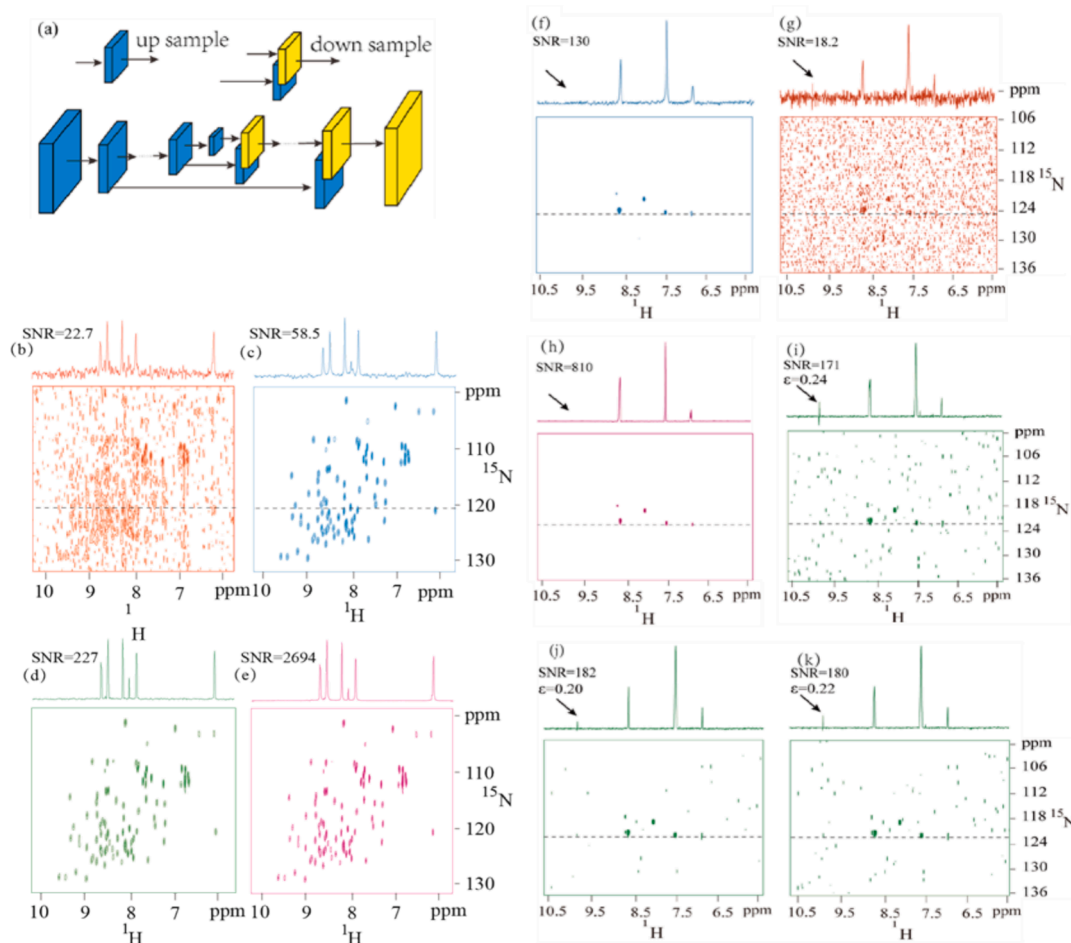


Fig. 9. (a) DN-Unet structure. (b–e) 2D ^1H - ^{15}N HSQC spectra of ubiquitin, (b) noisy spectrum, (c) reference spectrum, (d) spectrum denoised by NASR [7], and (e) spectrum after DN-Unet denoising, (f–k) corresponding ^{15}N - ^1H planes of 3D HNCO spectra of azurin. (f) Reference spectrum, (g) noisy spectrum, (h) spectrum after DN-Unet denoising, (i–k) spectra denoised by NASR using different parameter values. Reproduced with permission from [74].

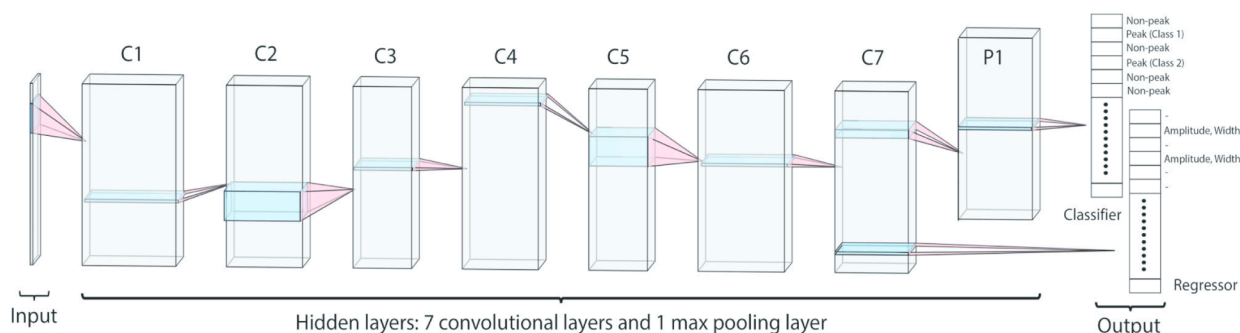


Fig. 10. The network framework of DEEP Picker. Reproduced with permission from [84].

showcasing its remarkable universality. Furthermore, DN-Unet excels in removing noise with high intensity, preserving weak spectral peaks, and significantly improving the SNR of 1D, 2D, and 3D NMR spectra. As shown in Fig. 9, the denoising effect of DN-Unet surpasses the traditional NASR method [7] by a significant margin. In addition, lightweight neural networks can still achieve outstanding results. Zhan et al. [75] introduced a Lightweight Denoising Network (LD-Net) to suppress noise and spurious peaks while recovering nearly completely submerged weak peaks in noise, thereby significantly improving SNR. Moreover, combining DL with traditional algorithms is also a viable approach. Zou et al. [76] proposed the Total Variation Conditional denoising Neural network (TVCondNet), which integrates the results of Total Variation (TV) denoising with the original noisy spectra as inputs, achieving noise prediction.

However, there are still some problems in the field of NMR denoising. When the noise is significantly larger than some of the weak peaks, DL often removes weak peaks together with the noise. At present, there is no good method to achieve the denoising of very weak peaks using DL. In addition, due to the influence of noise, the peak intensity in the denoised spectrum is closer to that in the noisy spectrum, deviating from the true one. There is still no effective solution for the restoration of peak intensity in denoising tasks.

4.3. Peak picking (Deconvolution)

Peak picking plays a crucial role in analyzing complex NMR spectra of proteins, RNA, DNA, and molecular complexes and mixtures. Many non-DL peak picking algorithms have been proposed and used, such as CcpNmr analysis (CCPN) [77] and PICKY [78]. However, these algorithms still perform imperfectly in highly crowded spectra. The introduction of DL offers a more efficient solution to the peak picking problem, achieving excellent performance for many challenging spectra.

In 2018, Klukowski et al. [79] proposed the NMRNet, successfully using a CNN to achieve peak picking. NMRNet was compared to previously available peak picking routines (CV-Peak picker [80], CCPN [77], and NMRViewJ [81]) in terms of manual annotation accuracy, resonance assignment completeness, and analysis of the CCL2 protein-glycan complex. The results showed that it significantly outperformed these methods when dealing with complex spectra that have overlapping signals and low signal-to-noise ratios. The advantage of the NMRNet may come from utilizing the whole peak neighborhood in the inference process by CNN. However, NMRNet's architecture is quite simple, containing only two convolutional layers and one fully connected layer. In subsequent studies, researchers achieved much better results with DL-based methods through careful design. For example, Schmid et al. [82] used inception layers and LSTM to perform peak picking in 1D NMR spectra, achieving results comparable to those of experts on several challenging spectra. The DEEP Picker proposed by Li et al. [83,84] not only achieves excellent performance in 1D spectra but also handles complex 2D spectra effectively. As shown in Fig. 10 (a), DEEP Picker consists of two branches. One of the branches is a classifier. The

extracted features are input into the classifier to classify the input point into one of three classes. The first class of peaks comprises single peaks, or the dominant peak among overlapping peaks, the second class of peaks comprises shoulder peaks among overlapping peaks, and the third class comprises non-peaks. The other branch of regression is used to estimate the position, amplitude, width of peaks, and the Lorentzian fraction of peak shape.

Currently, these peak-picking methods are all performed on Fourier-transformed spectra, but for serial NMR data sets, Radon Transforms (RT) are the most commonly used processing method, providing sensitivity and resolution gains. However, the spectral line shape after Radon transformation is not Lorentzian, so previous peak picking methods cannot be directly used. Nawrocka et al. [85] trained a U-Net network with synthetic 2D Radon spectra to perform peak picking on 2D Radon spectra. Radon transforms and neural network-based peak-picking methods can be used to characterize metabolomic mixtures and significantly reduce the experimental time required for measurements. To make it easier for researchers to use, Nawrocka et al. also developed a user-friendly graphical interface, which is crucial because not all researchers have a deep learning background. In addition, Klukowski et al. [86] developed the NMRtist online platform, based on ARTINA [87], which includes peak picking capabilities. The main components of ARTINA are ResNet and GNN. NMRtist encapsulates the complex workflows of ARTINA, allowing users to perform various spectral analysis tasks without the need to set parameters. In addition to peak picking, NMRtist can also perform functions such as protein structure determination (see Section 4.4.1 for details).

4.4. NMR parameter and chemical structure inference

NMR parameter and chemical structure inference plays a crucial role in various applications such as structure validation, automatic assignment, and compound elucidation. Neural network technology has been extensively used for inferring NMR spectra from chemical structure, or inferring NMR parameters and chemical structure from NMR spectra. Initially, it was applied to specific groups of organic compounds [88,89], and later extended to encompass different types of organic molecules with various nuclides [90,91].

NMR measurements yield information on chemical shifts, signal intensity, and spin-spin coupling constants. The chemical shift can indicate the chemical environment of the nucleus, and the Scalar Coupling Constant (SCC) conveys structural and stereochemical information. Resolving the molecular structure from the measured NMR spectrum is an inverse problem, involving the retrieval of 'hidden' information from the observed data. To obtain a unique solution, certain constraints must be imposed. Additionally, supplementary knowledge is often necessary to establish a coherent structure from the obtained information.

Currently, quantum chemical calculations, typically based on Density Functional Theory (DFT) methods, are the established approach for calculating NMR parameters for use in 3D structure verification [92]. Although the DFT method has high accuracy, it takes from several

minutes to days to perform a calculation on a single molecule. If exploring the conformational space, it will require still more time. In contrast, empirical methods rely on extensive data collection. The commonly used approach involves searching the atomic structure environment database (represented by HOSE codes) [93] and then inferring the chemical shifts of the most similar atoms. The average deviation between the inferred and actual values of ^1H chemical shifts is typically around 0.2 to 0.3 ppm. However, the original form of HOSE codes does not account for stereochemistry and cannot differentiate between conformers. Furthermore, the accuracy of database inference relies heavily on the quality of the database and the similarity between new and known HOSE codes.

Compared to purely database-based inference, ML can utilize some easily accessible molecular information to infer chemical shifts on a large scale, reducing the inference time by several orders of magnitude. Paruzzo et al. [94] developed ShiftML, based on local environment-ML, to infer chemical shifts of molecular solids, which yields accuracy levels for ^1H similar to those of DFT. However, other nuclei exhibit greater errors compared to DFT. The neural network method proposed by Aires-de-Sousa et al. [91] can be used for the analysis of relative 3D configuration, but cannot deal with 3D conformational analysis. Gerrard et al. [95,96] developed a system named IMPRESSION to infer NMR parameters for organic molecules, including 3D discrimination. This method achieves high accuracy, with errors mainly arising from the underlying DFT method and the chemical space range covered by the training set. However, it requires a geometry as input, which can be challenging to determine. Li et al. [97] developed a protein chemical shift predictor, which almost reached the accuracy limit of the method of extracting features from the structure. Jia et al. [98] developed SRCV, based on ML, to identify and extract information from processed spectra, including ^{13}C and ^1H chemical shifts and their integrals.

Regarding the inference of coupling constants, Binev et al. [99] made an early attempt using associative neural networks integrated with multilayer perception and k-nearest neighboring. Their method infers ^1H - ^1H coupling constants with a Mean Absolute Error (MAE) of 0.6–0.8 Hz. However, this approach utilizes two-dimensional (2D) vectors as input, which cannot adequately represent the 3D structure of molecules. Consequently, there are limitations in accurately determining complex

Table 3

Technical details of different DL algorithms for chemical structure prediction.

Authors	Network Foundation	Database	Evaluation Index	Comparison method*
Pons et al. [139]	FCNN	BMRB	$p'(O)$	Previous work [161,162]
Huang et al. [141]	CNN	GDB-13 & MestReNova & HMDB	PRC-AUC & Micro-average F1 & Accuracy	HMDB
Kuhn et al. [142]	CNN	nmrshiftdb2 & BMRB	Accuracy & F1 & Recall et al.	MobileNetV2 & ORBCompare & SSICompare et al.
Kim et al. [143]	CNN	CHNMR-NP & NPATLAS & NPASS et al.	Precision & recall & F1	SMART 2.0 [144] & NMRShiftDB
Miao et al. [146]	LSTM [45]	DSSP & BMRB	Accuracy	CSI2.0 & CSI3.0
Zhang et al. [147]	PNN	BMRB & PDB	Accuracy	Random Forest Classifier & 1D convolutional classifier
Devata et al. [148]	GNN [48]	QM9	Accuracy	QM9
Klukowski et al. [87]	ResNet & GNN [48]	BMRB & PDB	RMSD	PDB [87]
Wetton et al. [149]	ResNet & GNN [48]	BMRB & PDB	Accuracy	CST [163]
Li et al. [153]	AlphaFold2 & PSVS	BMRB & PDB	GDT-TS & RPF-DP& RDC et al.	AlphaFold2 [150]
Ma et al. [159]	AlphaFold2	PDB	pLDDT & S2	AlphaFold2 [150]
Klukowski et al. [160]	ResNet & GNN [48] & AlphaFold	BMRB & PDB	Accuracy & RMSD	experimental NMR data BMRB & PDB

* Comparison method is the method with which the deep learning-based method in the first column of the table is compared in terms of performance.

Table 2

Technical details of different DL NMR parameter inference algorithms.

Authors	Prediction parameter	Network Foundation	Database	Evaluation Index	Comparison Method*
Liu et al. [105]	chemical shift	DenseNet [66]	CSD	RMSE	KRR
Hack et al. [106]	chemical shift	GNN [48]	Ilm-NMR-P31	MAE	HOSE & linear increment models
Jonas et al. [107]	chemical shift	GNN [48]	NMRShiftDB2	MAE & RMSE	HOSE & DFT
Guan et al. [108]	chemical shift	GNN [48]	NMRShiftDB	MAE & RMSE	DFT
Yang et al. [109]	chemical shift	GNN [48]	RefDB & SHIFTX human & metabolome 4.0	RMSE	SHIFTX+ & UCBSHIFTX & CS2Backbone et al.
Han et al. [113]	chemical shift	GNN [48]	NMRShiftDB2	MAE & RMSE	FCG & SG-Only SG-IMP & SG-IR
Ai et al. [114]	chemical shift	GNN [48]	Exp5K & AIST	R^2 & MAE & RMS et al.	CASCADE & Ensemble NMR Prediction
Kaneko et al. [115]	shielding constant	GNN [48]	S-BLDA	RMSE	DFT
El Samman et al. [116]	chemical shift	GNN [48]	QM9 & NMRShiftDB2 & NPMRD	MAE & RMSE	Previous work [134]
Rull et al. [117]	chemical shift	GNN [48]	NMRShiftDB2	MAE & RMSE & SD	HOSE & CDK
Chen et al. [118]	chemical shift	GNN [48]	Glycosciences.DB & CSDB & PDB	RMSE	SchNet & DimeNet++ & ComENet & SphereNet
B��nkstad et al. [119]	chemical shift	GNN [48]	CASPER	MAE & RMSE	DFT
Han et al. [122]	chemical shift	GNN [48] & DFT	NMRShiftDB2	MAE & RMSE	HOSE & GCN
Xu et al. [123]	chemical shift	transformers [50]	NMRShiftDB2 & QM9-NMR & CSD & Aflow & Materials Project	MAE & RMSE & R^2	DFT
Jian et al. [132]	SCC	GNN [48] & transformers [50]	CHAMPS	MAE & logMAE & MAPE	DFT
Fang et al. [133]	SCC	GNN [48]	QM9	logMAE	DFT

* Comparison method is the method with which the deep learning-based method in the first column of the table is compared in terms of performance.

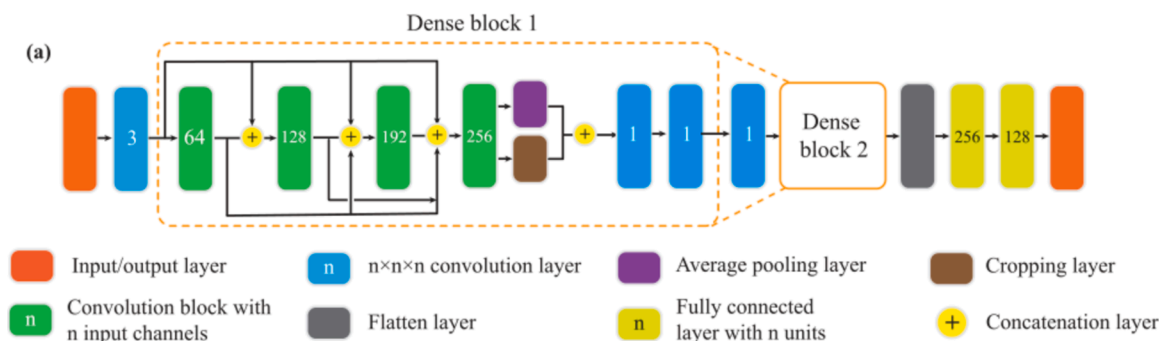


Fig. 11. The network flow chart of MR-3D-DenseNet. Reproduced with permission from [105].

molecular structures. Shibata and Kaneko [100] realized a Root-Mean-Squared Error (RMSE) range of 0.4–1.8 Hz through LightGBM, using RDKit descriptors and decision trees. However, compared with DFT, the accuracy is lower. Ito et al. [101,102] proposed a method that combines Quantum Mechanics (QM) and ML to infer ^1H - ^1H coupling constants and ^1H chemical shifts of components of mixtures, and thus 2D J -resolved spectra of mixtures. The accuracy of inferred coupling constants was assessed with validated data, resulting in a RMSE of 1.2 Hz [101]. Gerrard et al. [95] employed the Coulomb matrix to encode and utilized Kernel Ridge Regression (KRR) to infer $^1J_{\text{CH}}$. They achieve an MAE of 2.0 Hz, which is equivalent to the DFT method. However, this method is only applicable to one type of coupling, $^1J_{\text{CH}}$, which is far from enough to be useful.

In addition, early ML methods often relied on feature engineering,

which could be influenced by human bias and incompleteness. They also typically required separate training for different atom types. In comparison, DL possesses self-learning and generalization capabilities. It can identify the most appropriate features during the training process without any manual intervention, enabling end-to-end functionality. Initially, feedforward networks were predominantly used to infer chemical shifts. Bohr et al. [103] employed a perceptron-class neural network to infer protein secondary structure, and Anker et al. [89] trained a fully connected neural network to infer ^{13}C chemical shifts with a MAE of 1.4 ppm, surpassing the performance of linear regression analysis. However, the computing capacity was limited during that period, impeding the advancement of accuracy.

As computing capacity has been continuously improving in recent years, DL performance has also advanced, reaching the accuracy limit of

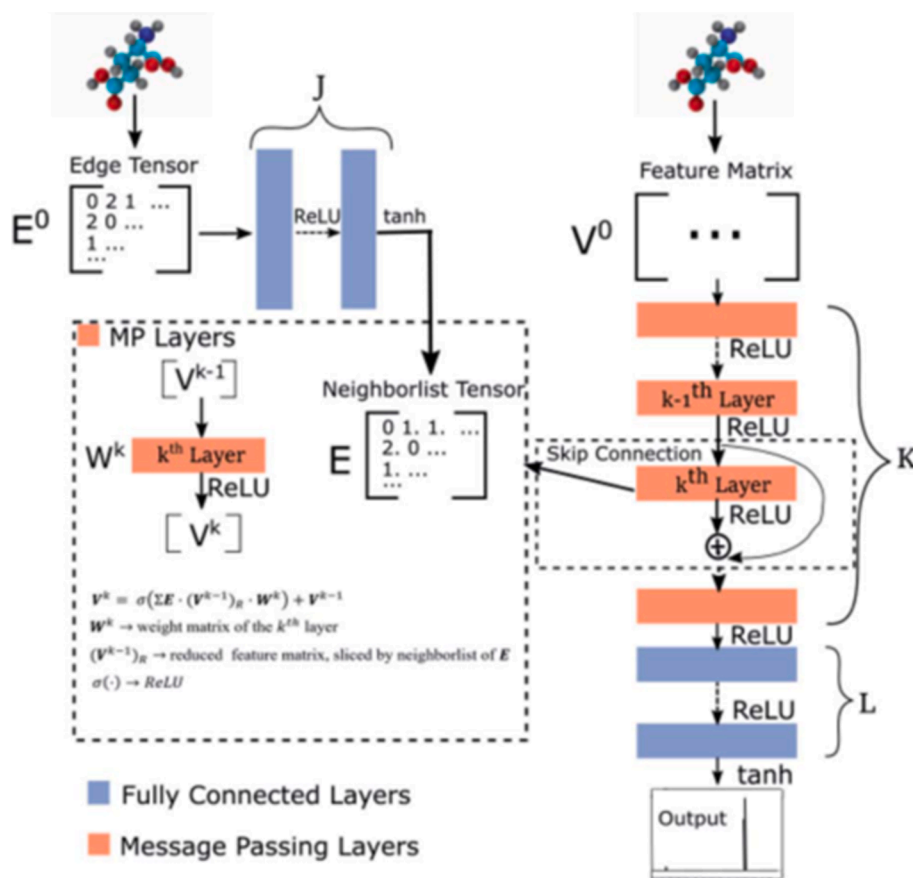


Fig. 12. Graph neural network architecture. K and L are the number of MP layers and fully connected (FC) layers, respectively. Reproduced with permission from [109].

secondary structure inference [104]. DL is now widely utilized in inferring NMR spectra from chemical structure or inferring NMR parameters and chemical structure from NMR spectra. We will introduce the applications of DL in predicting NMR parameters and chemical structures in sections 4.4.1 and 4.4.2, respectively, and summarize them in Tables 2 and 3.

4.4.1. NMR parameters

Liu et al. [105] developed MR-3D-DenseNet, a DL approach to inferring chemical shifts. The training data for MR-3D-DenseNet are obtained from the Cambridge Structural Database (CSD). Fig. 11 shows the architecture of MR-3D-DenseNet, which consists of a convolutional layer, two DenseNet blocks, and a transition convolutional layer. The output from Dense block 2 is connected to a fully connected layer with 256 neurons, then to a 128-neuron layer, and finally to the output layer. Dense block consists of four repeating units, where each unit comprises two bottleneck convolutional layers, followed by the convolutional layer with 64 channels. Two convolutional layers are used after concatenating cropping and pooling at the end of each block. Compared with the KRR approach [94], MR-3D-DenseNet exhibits better inference performance. However, due to the limited training data, it cannot attain the same accuracy as the best ab initio methods. It is plausible that the multi-resolution architecture employed in MR-3D-DenseNet can be utilized to infer other chemical properties that are influenced by different length scales.

Compared to the CNN-based network, the GNN is a completely new network that more easily learns molecular structure information. GNNs leverage the graph structure to simulate the topological arrangement of diverse molecules. In addition, self-attention excels at handling intricate macromolecular structures. The combination of GNNs and the self-attention mechanism enables comprehensive learning of molecular structure information. Hack et al. [106] established an open ^{31}P chemical shift database named Ilm-NMR-P31 with nearly 15,000 entries, filling the current gap in open resources. Based on this database, they utilized models based on different algorithms for inferring chemical shifts. Among these models, the trained GNN-based model achieved the highest accuracy, further demonstrating the advantages of DL.

Jonas et al. [107] proposed a GNN-based network to infer ^1H and ^{13}C chemical shifts and uncertainties. This marked the first application of a GNN to infer chemical shift. This approach outperforms classical methods based on HOSE codes, and has comparable reliability to inference based on DFT for the specific molecules studied. However, due to the limitations of the training data set, the method overlooks solvent and temperature effects, which can influence chemical shift values in experiments. Furthermore, it disregards stereochemical and geometry-specific effects. Integrating geometric characteristics into the model may further enhance accuracy. Guan et al. [108] also developed a GNN method using 3D structures for inferring ^{13}C and ^1H chemical shifts. They used three subsets of NMRShiftDB: NMR8K, DFT8K, and Exp5K. Three separate GNNs, named DFTNN, ExpNN-dft, and ExpNN-ff respectively, were trained with the same network structure hyperparameters. DFTNN was trained by DFT-calculated chemical shifts from DFT8K. Subsequently, the model was retrained using the chemical shifts from Exp5K while retaining the DFT geometries, resulting in ExpNN-dft. Lastly, the model was retrained by chemical shifts from Exp5K, but with Merck molecular force field structures replacing the geometries, leading to ExpNN-ff.

Yang et al. [109] also presented a method based on GNNs to accurately simulate chemical shifts and infer shifts of organic molecules. As shown in Fig. 12, the network is composed of a density network, a message passing neural network, and a dense network with chemical shifts as the output. The input for this model is the molecular graph, and the output is a set of inferred chemical shifts. The training data of the model come from RefDB [110], SHIFTX [111], and the Human Metabolome database [112]. Compared to the previous method, the network is more efficient and can be applied to any input molecular type.

Additionally, the network does not rely on feature engineering and achieves comparable results to other DL methods despite using a smaller dataset. If additional parameters like distance information or the identities of nearby atoms are provided, the accuracy will improve. However, this method should balance the number of inferred parameters and accuracy.

GNNs have limitations due to high space complexity and computational issues, restricting them to small molecules. Han et al. [113] proposed an approach to solve this issue by introducing a scalable GNN. This approach treats heavy atoms and their chemical bonds as nodes and edges to make the graph representation of a molecule sparser, effectively reducing space complexity. The network utilizes the attention mechanism and residual connection to improve node embedding vectors, and adds node and graph embeddings as the readout function for node-wise chemical shift inference. This method has achieved accuracies of 1.43 ppm and 0.28 ppm for ^{13}C and ^1H , respectively. The performance of the network can be further improved if the 3D geometric information of molecules is used as additional input. However, acquiring geometric information is expensive and it is not always available.

With the growing need for precision, more intricate versions of GNNs have been utilized. Ai et al. [114] developed a GFN2NMR method based on a deep graph convolutional network to calculate ^{13}C NMR chemical shifts. The performance of GFN2NMR is comparable to that of DFT-based methods, and the computational efficiency is greatly improved. This method resolved the controversial J/K rings junction problem of maitotoxin, and provides a new tool for the structural verification and assignment of large structures. Kaneko et al. [115] also used the GCN-based DL model SchNet to infer the NMR shielding constants of Na clusters and structures. The RMSE of the trained model was 35.2 ppm, which is basically the same as the RMSE of the Lasso regression model [115].

However, such graph-based neural networks require large databases. If the training dataset is small, the accuracy of the network will be low. To circumvent this problem, El-Samman et al. [116] adopted transfer learning, which utilizes existing generalizable knowledge within pre-trained graph-based molecular representations to effectively retrain and perform well on smaller datasets. They explored transfer learning methods from the pretrained network SchNet to infer atomistic properties and molecular properties like ^{13}C chemical shifts, pK₁, and logS. The method uses a relatively small training set (<1000 training points). Although the accuracy is not perfect, it is better than that of other networks using the same number of training sets. Rull et al. [117] proposed a graph neural network for inferring ^{19}F shifts and solvent-specific ^{13}C shifts. The network was trained using only 1000 data points, with a MAE less than 5 ppm. However, the trained network model is restricted to small organic molecules in solution. From a comprehensive perspective, the performance of small datasets is still inferior to that of large datasets. The use of simulated data may be a feasible approach.

The interpretation and characterization of carbohydrate-specific data are usually more complicated than for protein data. Chen et al. [118] proposed GlycoNMR, a large-scale carbohydrate NMR dataset specifically curated for ML, containing carbohydrate structures and assigned chemical shifts. They also developed a baseline 2D GNN method, based on GlycoNMR, for inferring the chemical shift of carbohydrates, with accuracy consistent with that of most advanced models, explaining the potential use of molecular representation learning in glycoscience research. B  nkestad et al. [119] designed GeqShift, based on E(3) equivariant GNN and 3D molecular structure, to infer chemical shifts of carbohydrates and further improve accuracy and robustness. The training dataset of GeqShift is derived from CASPER [120]. The input required contains atom types, neighboring hydrogen information, bond types, and relative distances between connected nodes. After passing through several equivariant layers, an invariant vector is generated. Subsequently, an MLP is employed to generate an invariant chemical shift inference. In comparison to models relying solely on 2D molecular structure information, GeqShift demonstrates a more than

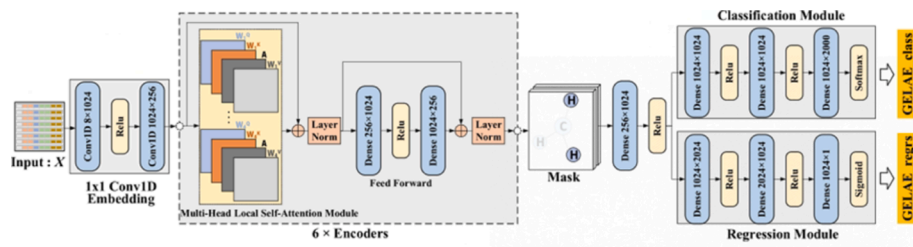


Fig. 13. Structure diagram of the GELAE model. Reproduced with permission from [132]

threefold reduction in MAE. Li et al. [121] designed a GNN-based network for inferring ^{19}F NMR chemical shifts. The training dataset comes from NMRShiftDB2. The model succeeds in incorporating the influence of paramagnetic shielding for ^{19}F and achieves an MAE of 3.6 ppm. This method has helped to correct the previously determined structure of fluorine-containing reagents and will provide an effective reference for determining the structures of organic fluorine compounds. Recently, Han et al. [122] designed a 3D GNN model combined with DFT-calculated NMR shielding tensor descriptors to infer ^1H and ^{13}C chemical shifts for small molecules. The trained model achieved an MAE of 0.50 ppm for ^{13}C and 0.08 ppm for ^1H in the NMRShiftDB2 and CHESHIRE datasets. However, the current input to the model is only one 3D structure per molecule. Providing multiple conformers, to compute the Boltzmann-weighted average prediction, could improve the reliability and versatility of the model.

Xu et al. [123] designed NMRNet, a Transformer-based network that utilizes a pre-training and fine-tuning approach to accurately infer ^1H , ^{13}C , ^{19}F , ^{11}B , or ^{17}O chemical shifts from molecular structures, and is the first model to integrate chemical shift inferences for both solid-state and liquid-state NMR. During pre-training, NMRNet learned general representations and knowledge of crystal structure data. For liquid-state NMR, weights pre-trained by Uni-Mol [124] were used directly, while for solid-state NMR, large-scale data from Aflow [125], CSD [126], and the Materials Project [127] were used to pre-train. After pre-training, fine-tuning can adapt the knowledge learned by pre-training to specific tasks (e.g. infer all isotope chemical shifts or individual isotope chemical shifts), reducing computational costs and improving the trained model's specificity on target tasks. NMRNet can also be used for structural assignments and stereochemistry determination. It may be possible to improve the model's accuracy by incorporating experimental factors such as solvent and temperature, and/or to extend it by including coupling effects.

Recently, more and more researchers have used neural networks to infer chemical shifts. Chen et al. [128] designed GT-NMR, a Transformer-based method that utilizes 2D molecular graph representations to infer ^1H and ^{13}C chemical shifts. Additionally, they proposed the size-normalized spatial score (nSPS) for evaluating the reliability of graph-based methods regarding molecular complexity for the first time. Williamson et al. [129] developed message-passing neural networks to infer chemical shifts in ^{13}C NMR spectroscopy. Prestegard et al. [130] used neural networks to infer ^1H chemical shifts and ^1H - ^1H scalar couplings from J -resolved spectra of systems with strong coupling spins at low field, but encountered some challenges.

As for identifying coupling constants, Karpov et al. [131] transformed molecular structures into 1D sequences and employed a Transformer for inference for the first time. However, this approach fails to preserve the 3D structural information for molecules. To solve the problem, Jian et al. [132] utilized unordered sets of chemical bonds to represent molecular structures, and developed a model named GELAE to efficiently and accurately calculate SCCs. The data used for training are sourced from the CHAMPS dataset. Fig. 13 illustrates the architecture of the GELAE model. The input is passed through a 1D convolution, with each input channel corresponding to eight chemical bond features. Next,

six encoders are employed to extract the interaction between chemical bonds. After encoding, the outputs are weighted by a mask, summed, and then fed into a fully connected network for inference. The output includes a classification model and a regression model. This network employs graph-embedding local self-attention modules instead of the global attention model, and uses multi-layer perceptual attention to accelerate model convergence and improve accuracy. Additionally, the use of a classification-based approach to solving the regression problem, which is a novel idea, has yielded promising results.

Fang et al. [133] introduced a network model called GAANN for inferring SCCs. GAANN can simulate molecule-like topological structures and accurately infer molecular properties, similar to DFT. Unlike previous GNN models, GAANN incorporates prior knowledge of angles. The network mainly comprises an encoder and a decoder. The encoder consists of a bond message-passing layer, a scalar coupling message-passing layer, and a self-attention neural network. The bond message-passing layer incorporates bond angle attention, and the scalar coupling message-passing layer incorporates angle features. Through the above message-passing layers, a self-attention network is used for extracting features. Finally, a multi-layer perceptron is employed for decoding and inferring SCCs. The test dataset used was a subset of the QM9 dataset. Compared to traditional attention mechanisms, the self-attention mechanism used in GAANN can handle long-distance and multi-level dependency relationships in complicated macromolecular structures. By incorporating more prior knowledge, the accuracy of the trained model can be further improved.

4.4.2. Chemical structure

The molecular structure determines the physical, chemical, and biological properties of a compound. Therefore, structure elucidation is complex and crucial. Traditionally, secondary structure is determined from distance information between nuclei obtained from NOESY experiments. NMR chemical shifts involve abundant structural information and are more easily obtained. The first method for inferring protein secondary structure from NMR data, CSI [135], was based on chemical shifts. Its accuracy could be further improved by incorporating machine learning, which is represented by CSI 2.0 [136] and TALOS-N [137]. However, most methods only offer a 3-state classification, which is inadequate. CSI 3.0 integrates models, including CSI 2.0 and TALOS-N, to further improve the inference performance of proteins [138]. In recent years, new methods have been continuously proposed to improve the accuracy.

Pons et al. [139] presented RESCUE, an assignment tool that utilizes artificial neural network technology to identify amino acid types based on observed chemical shift values in the ^1H spectrum. This approach exhibits a higher success rate compared to previous methods. However, the reported level of errors in this work is influenced by potential problems including assignment errors and even primary structure errors in the Biological Magnetic Resonance Data Bank (BMRB) itself [140]. It is anticipated that as the accuracy and scale of the database continue to improve, the performance of RESCUE will get better.

CNN remains one of the most commonly used networks for chemical structure inference. Huang et al. [141] designed a network based on

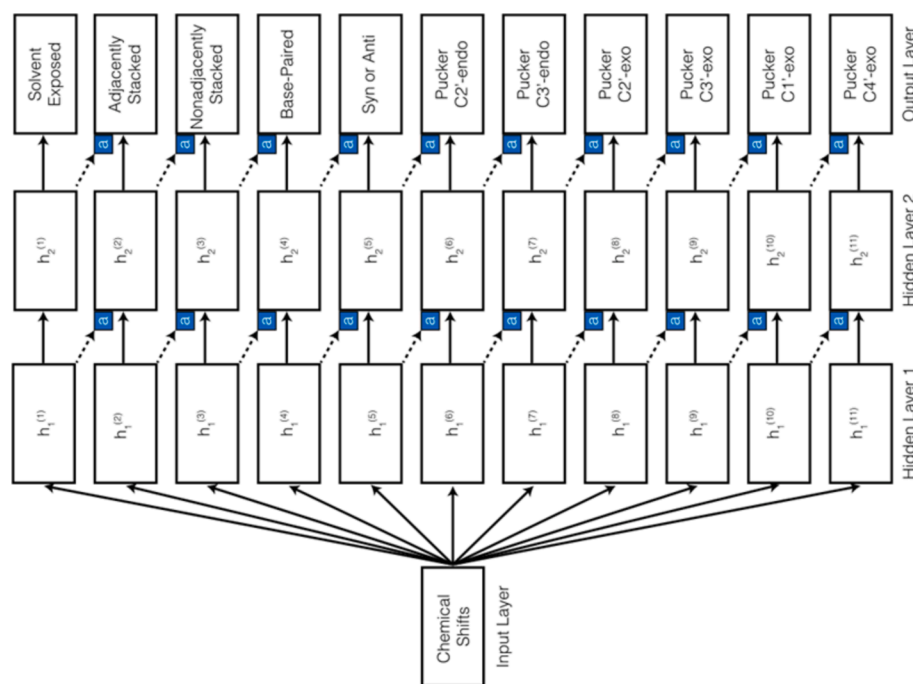


Fig. 14. Architecture of CS-Annotate. $h_i(j)$ stands for the i -th hidden layer for the j -th task. Reproduced with permission from [147].

CNN to quantitatively rank the most probable structural connectivities of unknown compounds. The input to the model comprises 1D ^1H and/or ^{13}C NMR spectra of small molecules, which contain no more than 10 non-hydrogen atoms. The trained model accurately inferred the correct constitutional isomer as the highest-ranking inference in 67 % of the cases studied, and as one of the top ten inferences in 96 % of the cases. However, it cannot infer relative stereochemistry (diastereomers). Kuhn et al. [142] used CNN to classify chemical compounds directly from NMR spectra (HMBC and HSQC) of small molecules without structural analysis, which can help reduce the time spent in searching for good structure candidates. If distance metrics, clustering methods, and parameters are correctly combined, clustering may yield better results. Kim et al. [143] developed DeepSAT, a CNN-based system for the identification and structure assignment of small molecules. It utilizes ^1H - ^{13}C HSQC spectra to infer chemical fingerprints, molecular weights and molecular structure categories. Its performance surpasses that of SMART 2.0 [144] and it addresses the data limitations of SMART 1.0 [145] and 2.0, effectively aiding in identifying molecular structures.

The peptide chain is an ordered sequence that is well-suited to LSTM. Hence, Miao et al. [146] proposed a method CSI-LSTM that utilizes bidirectional Long Short-Term Memory (biLSTM) to infer the secondary structure (helix, strand, or coil) of proteins. The training sets were acquired from a dictionary of protein secondary structure (DSSP) and the BMRB. When compared to CSI 2.0 [136] and CSI 3.0 [138], CSI-LSTM benefits from a larger dataset and constructs a model that is more suitable for protein sequences. LSTM can effectively utilize protein sequence information, unlike SVM and MLP. As a result, CSI-LSTM improves accuracy. Additionally, unlike CSI 2.0 and TALOS-N [137], DL performs automatic relative offset correction, rendering CSI-LSTM unnecessary for chemical shift re-referencing.

Zhang et al. [147] developed CS-Annotate to infer eleven residue structural properties of RNA, including solvent exposure, based on nonexchangeable ^1H and ^{13}C chemical shifts of residues in RNAs. Fig. 14 shows the main architecture of CS-Annotate. During training, the model starts as a regular neural network with $h_1(1)$ and $h_2(1)$ as hidden layers for the first task (solvent exposure). For subsequent tasks (adjacently stacked), the parameters learned previously are fixed, and $h_2(2)$ takes

input from $h_1(2)$ and $h_1(1)$ through a lateral adapter layer. This parameter transfer continues for each remaining task. CS-Annotate surpasses common baseline models like independent multilayer perceptron classifiers [147] and recurrent neural network classifiers [147] in terms of average balance accuracy. Nevertheless, CS-Annotate also has limitations. Firstly, its inferences are significant only if assigned chemical shifts are available for some nonexchangeable ^1H and ^{13}C atoms in each residue of an RNA. Secondly, CS-Annotate's inference of residues as solvent-exposed relies on a threshold. A higher threshold (closer to the tail of the distribution) will lead to an imbalance in the target data for solvent accessible surface areas, impacting the classifier's robustness. Lastly, CS-Annotate is order-dependent, and identifying the optimal target order would necessitate exploring over 39 million combinations. Given limited computing power, no investigation was conducted into the effect of changing the order in which the different tasks of CS-Annotate are carried out.

Similar to inferring NMR parameters, the GNN is also adept at inferring chemical structures. Devata et al. [148] proposed DeepSPInN, a framework utilizing a Monte Carlo tree search and a GNN for multimodal analysis and inference of molecular structures using infrared and ^{13}C NMR spectra of small molecules. The training set for DeepSPInN is QM9. DeepSPInN is the first application of multimodal data in molecular structure inference, with an accuracy of 96 % on molecules containing fewer than 10 heavy atoms.

Klukowski et al. [87] proposed ARTINA for analyzing protein NMR data automatically. ARTINA utilizes protein sequences and NMR spectra as input to identify protein structures, cross-peak positions, chemical shift assignments, and distance restraints from NOESY spectra. As shown in Fig. 15, the process begins with visual spectrum analysis, and utilizes ResNet to identify cross-peak positions in NMR spectra. The coordinates of signals are then fed into the FLYA algorithm to provide initial chemical shift assignments. Subsequently, information from thousands of previously solved protein structures using NMR is incorporated through a deep GNN trained on BMRB/PDB (Protein Data Bank) deposits. This additional input allows the reassessment of cross-peak positions in the second FLYA call, completing the chemical shift refinement cycle. For structure refinement, ten variants of NOESY peak lists are

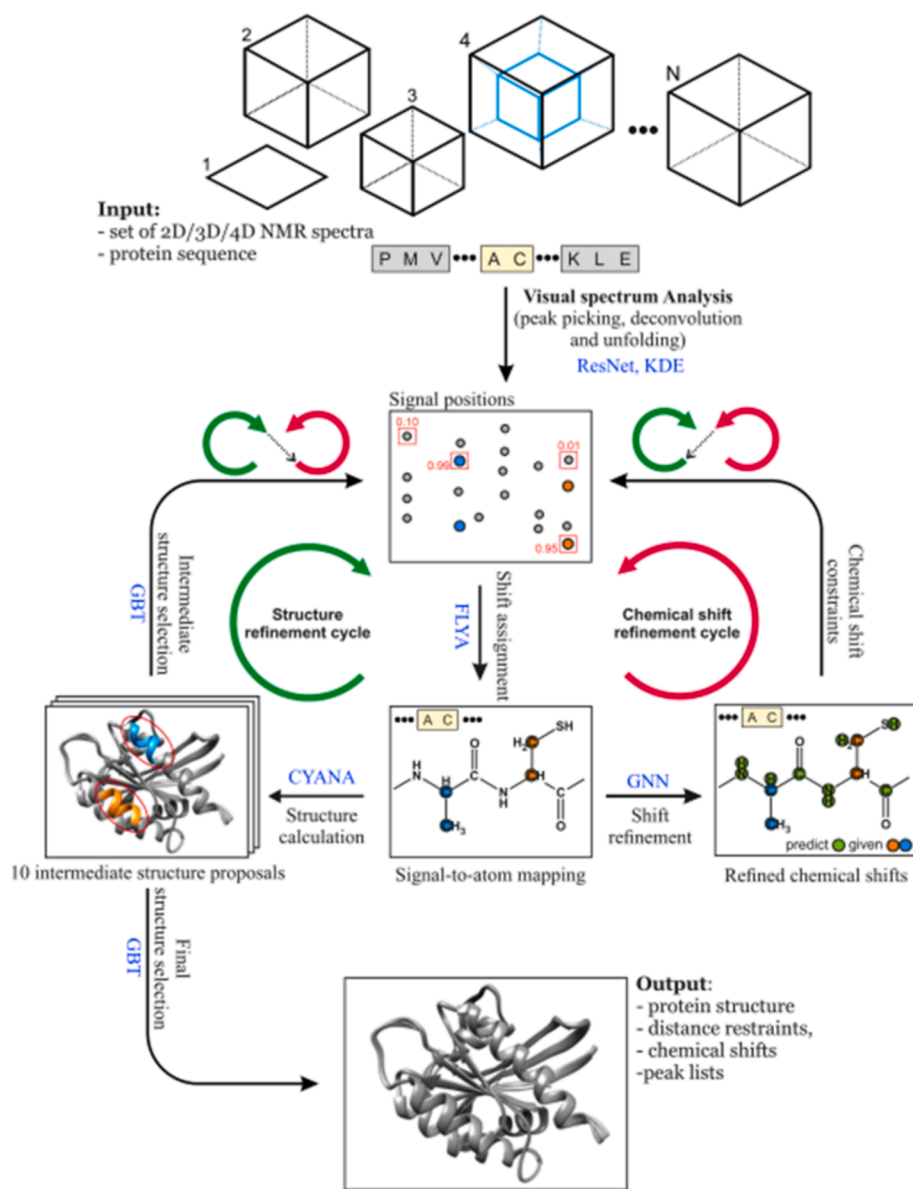


Fig. 15. The workflow of ARTINA.
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created by adjusting the confidence threshold of signal selection in ResNet. Every group of NOESY peak lists is used in the CYANA structure calculation, resulting in ten intermediate structure proposals ranked by the gradient boosted trees model. The best structure proposal then serves as contextual information, concluding the structure refinement cycle. In addition, they also combined CST and ARTINA to propose an automated workflow based on DL [149], allowing CST to be performed in a few minutes or hours of computational time (taking previously one to two weeks).

Compared to previous methods, ARTINA understands the whole workflow from spectra to structures instead of focusing on individual steps. In most cases, the accuracy is comparable to that of human experts. Inconsistencies mainly occur due to displacements of terminal secondary structure components or incorrect loop structures. ARTINA can also be utilized for deconvolution and resonance assignment. Recently, they developed an automated online platform called NMRtist [86], based on ARTINA, which can automatically analyze protein NMR spectra within a few hours. The functions include peak picking, assignment, and structure determination. NMRtist also offers virtual

storage, making it possible to obtain large-scale datasets. Future plans involve expanding the platform to handle protein–ligand complexes and RNA.

The introduction of the deep neural network AlphaFold [11] in 2020 and the improved version AlphaFold2 [150] (AF2) in 2021 and AlphaFold3 in 2024 [151] have completely changed the research approach of structural biology. AF2 can infer the atomic details of many proteins from amino-acid sequences at nearly experimental accuracy, outperforming X-ray crystallography and cryo-electron microscopy [152] in speed and convenience, especially in challenging cases. Li et al. [153] generated AF2 models to infer small monomer protein structures and assess how well these models fit experimental NMR data. Except for the presence of multiple conformational states of proteins [154] or errors in experimental structure determination, in most cases the trained AF2 models were almost identical to models deposited in the PDB based on experimental data. Although AlphaFold, AlphaFold2 and AlphaFold3 have achieved significant success, they also have some limitations, including predicting the impact of mutations on structure, protein aggregation, inferring the structure of protein–ligand and protein–protein

complexes, and treating inherently disordered proteins [155–158].

Due to the impressive performance of AF2, Ma et al. proposed a method called cdsAF2 [159], which employs 3D protein structures processed by AF2 to infer N–H S^2 order parameters of the protein backbone. The method can infer the secondary structures, loops, and tails of varied protein regions along the main chain, and provide the dynamic assignment of AF2 structures. However, if AF2 incorrectly infers protein regions with high confidence in structures, dynamic inferences based on structural dynamics models will fail. This may require the development of predictive factors to narrow this gap. Klukowski et al. [160] combined ARTINA with AlphaFold and UCBShift [97] to further improve accuracy and significantly reduce measurement time.

Generally speaking, DL depends heavily on the size, quality, and diversity of training data. Consequently, in the absence of experimental data, accuracy is essentially restricted by DFT methods, basis sets, and other computational errors. DL possesses strong generalization abilities, enabling models to infer situations that lie between known situations. However, if there is a significant deviation of the data to be processed from the training data, the DL algorithm may encounter challenges. Thus, it is unclear whether its advantages will be maintained when determining new structures, despite some papers reporting performance exceeding that of DFT.

Expanding the training set is the simplest method to enhance performance. However, current databases are very small compared to the vastness of chemical space. Utilizing synthetic data for training can address this issue, but then a trained model may not generalize well to real data. Furthermore, apart from improving individual models and algorithms, it is also possible to integrate different algorithms, like ML and HOSE codes, into a cohesive system to improve performance.

4.5. Compound identification and assignment in mixtures

NMR spectroscopy can provide detailed atomic-level information that is closely related to chemical structure, so it is widely used for identifying compounds and separately assigning them. However, when dealing with complex mixtures, identifying components from NMR spectra becomes very difficult. In recent years, some DL methods have been proposed that address the task of compound identification and

assignment in mixtures.

Different DL methods for compound identification and assignment in mixtures may require different NMR spectra with different dimensions. Some methods use 1D spectra as input, while others rely on 2D or 3D spectra. For example, Wei et al. [164] employed 1D ^1H NMR spectra and proposed an end-to-end DL method using a Pseudo-Siamese Convolutional Neural Network (pSCNN) to identify compounds in mixtures. A Siamese network is a kind of neural network structure usually used to solve tasks of similarity comparison, such as face recognition [165]. Similar data are simultaneously input into multiple subnetworks with the same structures and weights inside a Siamese network. The two parallel subnetworks used by Wei et al. have the same structures. However, since the weights of the two parallel subnetworks are trained separately, it is called a pseudo-Siamese network. The input of one subnetwork is a 1D NMR spectrum of a pure compound and the input of the other is a 1D NMR spectrum of a mixture. The pSCNN overcomes the issue that chemical shift of the same compound can vary between different mixtures. Even with chemical shift changes of up to 0.015 ppm, the pSCNN can usually still correctly identify components in mixtures. Wagner et al. [166] proposed an FCNN-based 'NMR fingerprinting' technique, with 1D ^{13}C NMR spectra as input, to automate the classification of structural groups in mixtures, yielding inferences consistent with true compositions. Li et al. [167] also employed 1D ^1H NMR spectra and used SVM, K-Nearest Neighbor (KNN), and RNN to achieve compound identification. Tian et al. [168] introduced a binary classification model, MatCS, to determine whether input NMR spectra, including both ^1H and ^{13}C NMR, are consistent with the input compound's molecular structure. Similarly, Zheng et al. [169] introduced CBMAFFNet (CNN BiLSTM Multi-step Attention Feature Fusion Network), which also utilizes ^{13}C and ^1H NMR spectra for the rapid classification of unknown psychoactive substances. Yang et al. [170] utilized 1D ^{13}C NMR spectra and proposed a system called Cross-modal Retrieval between Spectrum and Structure (CReSS) to match an unknown compound to a known molecular structure in a library. As shown in Fig. 16, CReSS contains an NMR encoder and a SMILES (Simplified Molecular-Input Line Entry System) encoder. The NMR encoder consists of CNN and MLP, and a transformer structure is adopted to construct the SMILES encoder. The ^{13}C NMR spectrum of an unknown compound is input into the NMR

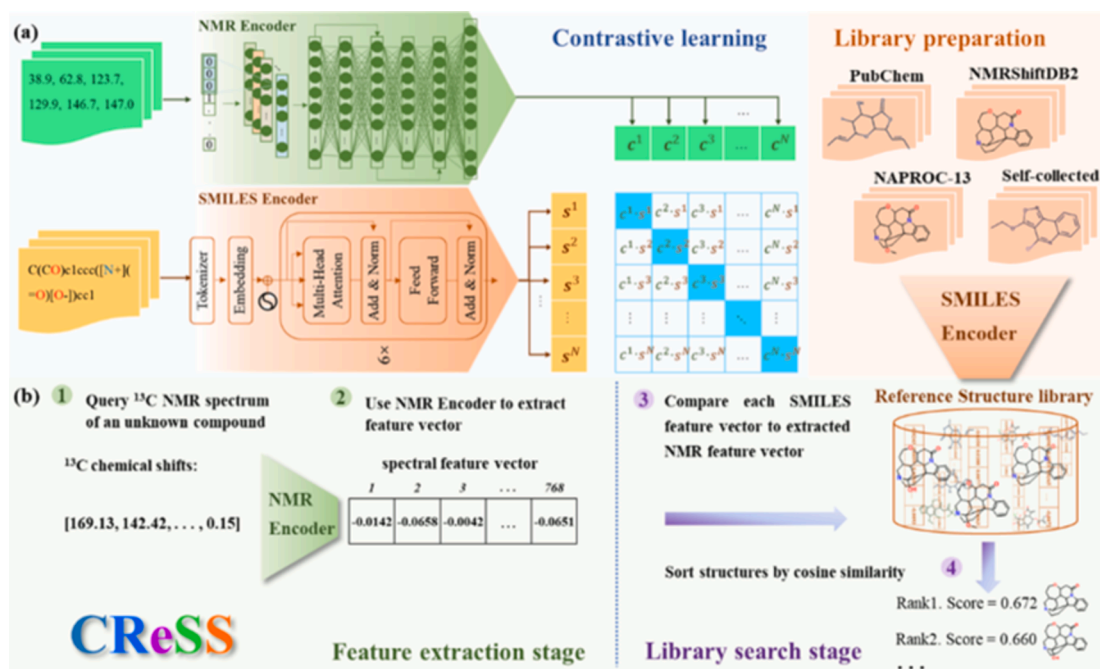


Fig. 16. (a) Workflow of the training model. (b) Workflow of CReSS. Reproduced with permission from [170].

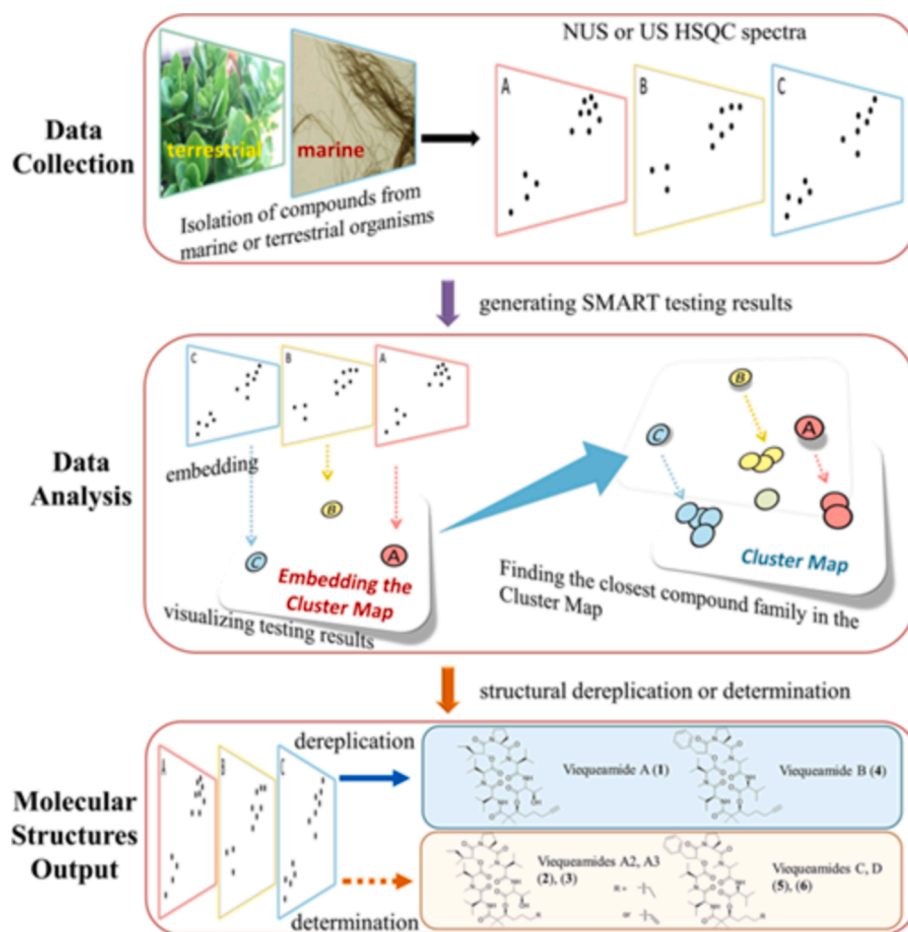


Fig. 17. Steps of the SMART method. Step one: Collect HSQC spectra of pure natural compounds. Step two: Map HSQC spectra into a cluster space. Molecules in the same natural product family are close to each other in the cluster space. Step three: Molecule structures are categorized according to their positions in cluster space. Reproduced with permission from [145].

encoder to extract the NMR feature vector, and ^{13}C NMR spectra of the known compounds are input into the SMILES encoder to extract the SMILES feature vector. The cosine similarity is then used to get the similarity between the NMR feature vector and the SMILES feature vector. The cosine similarity is calculated with

$$\text{Sim}(\mathbf{c}, \mathbf{s}) = \frac{\mathbf{c} \cdot \mathbf{s}}{|\mathbf{c}| |\mathbf{s}|} \quad (3)$$

where $\mathbf{c}$ and $\mathbf{s}$ are the NMR feature vector and SMILES feature vector respectively. The higher the similarity, the closer the spectrum of the unknown compound is to that of the known compound, and hence the more similar their structures are likely to be.

2D spectra are also often employed in component identification. For example, HSQC spectra are used in the SMART (Small Molecule Accurate Recognition Technology) method [145]. It was proposed by Zhang et al. in 2017, and developed based on CNN with a Siamese architecture and clustering (Fig. 17). The Siamese network maps HSQC spectra to a representational space in which similar HSQC spectra are mapped to close positions in the cluster space and spectra with significant differences are mapped to distant positions. The experimental results prove that the SMART method has the ability to recognise similarity between an unknown natural product and a known one and categorize it into the appropriate compound family. Kuhn et al. [171] proposed a DL method for identifying substructures in 2D NMR spectra of mixtures. They used both HSQC and HMBC spectra as input for the DL model, achieving better results than using HSQC spectra alone. Notably, even though they trained the model with pure compounds, the trained model could also

identify substructures in mixtures, demonstrating the robustness of this DL approach. Yuan et al. [172] presented the HSQCid method, which uses ^1H - ^{13}C HSQC spectra and known chemical structures to assess the similarity between unknown natural products and known compounds. In addition, a DL model using the 3D HNCA spectra of proteins as input was proposed by Rasulov et al. [173], and the backbones of common proteins (GB1, MBP, and INMT) were rapidly assigned.

Overall, DL has shown great performance in component identification. However, the types of spectra used as inputs in DL models are quite limited. If more types of spectra can be input to train the DL model, it is believed that DL will be able to tackle even more challenging tasks in component identification and separation.

4.6. NMR based metabolomics

In the early years, DL methods gained much attention in domains such as genomics, transcriptomics, proteomics, and imaging analysis, but not in metabolomics. In the traditional methods of determining the important variables (such as biomarkers) in the NMR spectra of different sample classes in metabolomics studies, samples were divided into two groups using Partial Least-Squares (PLS), the differences (especially important variables) between the two sample classes were compared, and then the importance of each variable was estimated using the Mean Decrease in Accuracy (MDA). The PLS method cannot analyze some cases that other ML methods, such as Random Forests (RF) [174] and SVM [17], can. In 2018, Date and Kikuchi [175] proposed DNN (Deep Neural Network)-MDA, combining DNN and MDA, for binary

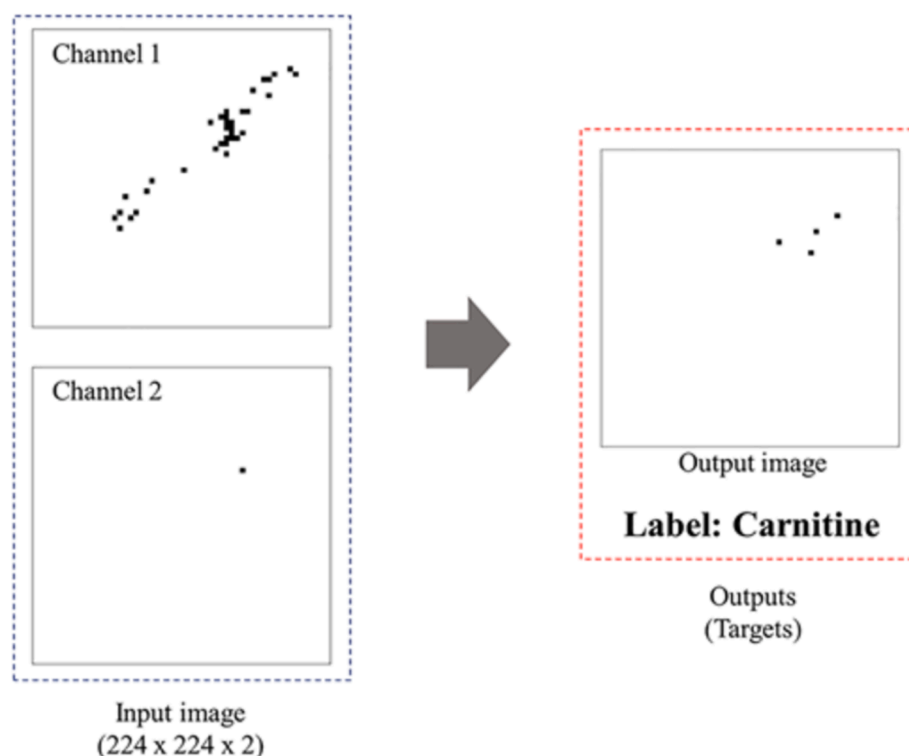


Fig. 18. The input and output of SMART-Miner. The input contains two channels: the ^1H - ^{13}C HSQC spectrum of a mixture and a single HSQC peak from one component of the mixture. Reproduced with permission from [180].

classification in supervised data analyses of metabolomics studies. Compared with PLS, SVM, and RF, DNN had higher accuracy when trained on a large sample database. However, in the case of a small sample database or highly biased data, DNN performed less well than SVM.

To extend the application range of DL, Asakura et al. [176] used a DNN-based method to solve regression problems in NMR-based metabolomics. They designed the EDNN method, which was built from multiple DNN classifiers and can predict the size of fish based on the metabolic profiles of their muscle. Compared to previous methods (DNN, random forest, and support vector machine algorithms), EDNN showed better regression performance for over half of the fish species. However, some results are worse, so DL methods for regression in metabolomics still have room for improvement.

In recent years, metabolomics has been applied in the field of microbial food safety. In some cases, classical microbiological detection methods, such as plating and qPCR analysis, are slow to obtain results. Wang et al. [177] proposed combining metabolomics techniques with DL algorithms to rapidly classify pathogenic and non-pathogenic microbes. The trained DNN model could identify the microbial strain from the metabolite profile with 99.2 % accuracy, using a basic feature selection method. However, the DL method was tested in a laboratory setting and could only identify a limited class of microbial strains in the dataset, needing further development to meet real-world requirements.

Reher et al. [178] proposed the application of an NMR-based DL tool, SMART 2.0, to analyze mixtures. SMART 2.0 was trained on HSQC spectra and mapped the spectra into a 180-dimensional embedding space by SqueezeNet [179]. They applied SMART 2.0 to an extract of filamentous marine cyanobacterium, aiding in the identification of a new chimeric macrolide, symplocolide A, and other relevant compounds.

Later, Kim et al. [180] proposed a DL method for the accurate identification of individual metabolites. They applied a U-Net [181] based CNN (SMART-Miner) to ^1H - ^{13}C HSQC NMR spectra, successfully

identifying peaks in complex mixtures, as depicted in Fig. 18. SMART-Miner showed both precision and recall exceeding 60 % for experimental spectra from open-source datasets, and its performance was competitive with that of other available tools. At present, one issue with SMART-Miner is the potential ambiguity in output when a query peak corresponds to multiple molecules.

Recently, Johnson et al. [182] designed a network based on MLP to quickly identify and quantify lipids, achieving accuracy comparable to Complete Reduction to Amplitude-Frequency Table (CRAFT). The network was trained on synthetic datasets. However, the network could only recognize lipid groups present in the training set, and was only validated under specific acquisition conditions. Transfer learning or diversifying the dataset may improve the generalization of the network.

The 1D NOESY-presat technique is commonly utilized for measuring metabolite spectra in biofluid samples like plasma or serum in metabolomics, but it cannot analyze macromolecular lipoproteins and small molecules separately. The 1D diffusion-edited method and the 1D Carr-Purcell-Meiboom-Gill (CPMG) method [183] can be used to separately analyze macromolecule lipoproteins and small molecule metabolites in biofluids. However, performing these two experiments on batch samples may be time-consuming. To solve this problem, Xiao et al. [184] developed a Unet-based neural network for processing 1D NOESY-presat spectra, separating the signals of macromolecules and small molecules. The statistical information yielded by Principal Component Analysis (PCA) [185] of macromolecule and small molecule signals extracted by the network is similar to that from analysis of 1D CPMG spectra and 1D diffusion-edited spectra. This method can save experimental time by using only 1D NOESY-presat spectra to analyse mixtures of macromolecules and small molecules without requiring 1D CPMG spectra or 1D diffusion-edited spectra, further demonstrating the potential of DL methods in metabolomics.

In metabolomics studies, DL has mainly been used for metabolite (molecule or biomarker) identification through classification and regression analysis, and has obtained high accuracy results. However, so

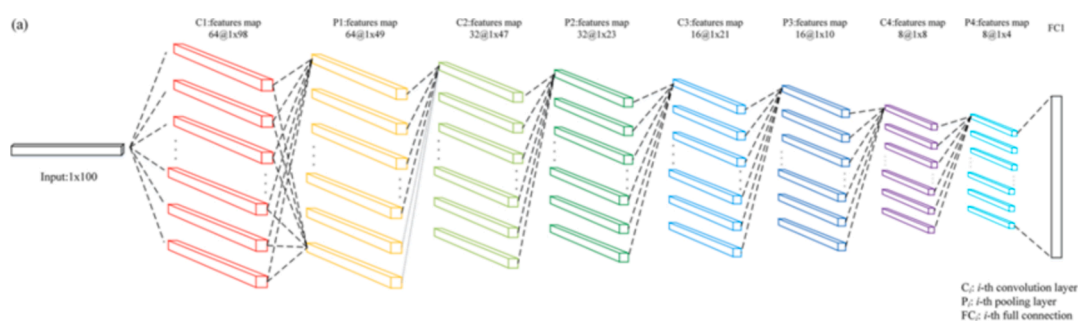


Fig. 19. 1D-CNN Structure.

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far, due to the limited sample numbers, complexity of the real data, and the simplicity of DL architectures, the performance gap between DL and traditional ML is not large, leading to a lack of wide application of DL in metabolomics. In addition, DL-based methods still need manual curation [186]. Building a more automated metabolomic analysis tool is a long way off.

4.7. Deep learning for low-field NMR

Low-field NMR mainly refers to the use of NMR instruments with low magnetic field intensity for analysis and detection. Because of their relatively low cost, small size, mobility, simple maintenance, ease of integration with other instruments, and other advantages, they are widely used in food, petroleum, medical, textile, and other industries. Many research directions, including online substance content measurement, can be explored using the data obtained from low-field NMR experiments.

The application of DL to low-field NMR methods can be divided into two areas. One is the use of post-processing of acquired NMR data to achieve results such as classification, image conversion, and prediction of various indicators [187–191]. The other is to apply DL techniques during experimental processes, for example in automatic tuning [192]. Most of the methods used are based on CNN and its variants.

Classification is a common research aim. In reference [191], Hou et al. employed a CNN to classify types of edible oils using data from a low field NMR instrument. To evaluate the influence of input dimensions on the performance of the CNN, Hou et al. trained two CNN models: a 1D-CNN, which took the transverse relaxation decay signal and a “ T_2 spectrum” fund by fitting the decay signal as inputs, and a 2D-CNN, which used 2D images of the transverse relaxation decay curve (signal as a function of time) and T_2 spectrum (signal as a function of T_2) as inputs. The results demonstratte that 1D-CNN (Fig. 19) had better classification ability for different types of edible oils.

The detection of moisture content is of great significance in the fields of chemical industry and food. Two main detection types are manual detection and online instrumental detection. However, manual detection methods, including density, distillation, and electrical separation, are time-consuming and laborious, and cannot meet the requirements of automation. Therefore, a detection method based on online low-field NMR data are attractive. The key to this method is to establish a relationship between the moisture content and the instrumentally measured data. Support Vector Machine (SVM) and Partial Least Squares (PLS) can deal with this problem, but the accuracy is not ideal. In reference [189], Yi et al. first used a convex region search method to extract the water peak from the NMR spectra of crude oil and then fed the processed data into a CNN-based classifier. The classifier separates the spectra into two categories: high moisture content and low moisture content. After classification, spectra from each category are input into two separate CNN-based regressors to infer the moisture content respectively. This approach enables accurately inferring the moisture content in crude oil, a task that is challenging for traditional methods.

Similarly, in reference [190], Zeng et al. established a set of prediction models combining a DL network with the Cuckoo Search (CS) optimization algorithm to estimate the moisture content of the sea cucumber. The CS optimization algorithm can extract significant features present in data to achieve dimension reduction, and effectively improve the performance of the trained model based on a small sample space and low class difference data. The authors first used the Chaotic-CS (CCS) algorithm to reduce the dimensions of HyperSpectral Imaging (HSI) data and Low-Field Nuclear Magnetic Resonance (LF-NMR) data, and then established two independent DL models and one Fusion Deep Learning (FDL) model. The results showed that both DL-based single-data detection and FDL-based multi-data detection are suitable for Moisture Content (MC) detection in sea cucumber pickling, benefitting the production process. The trained FDL model is more accurate and stable than the independent DL models, illustrating the advantages of joint multi-data detection.

DL has played a significant role in post-processing NMR data acquired at low field. In experimental methods for low-field NMR, DL has also found some applications. A study of T_2 data inversion benefitted from DL [192]. Luo et al. used an ATTention multi-scale Convolutional Neural Network (ATT-CNN) for the inversion of NMR relaxation data. Processed by the network, noisy echo data can become a T_2 spectrum. The results show that the trained ATT-CNN model better adapts to echo signals with lower SNRs and generates sparser and smoother T_2 spectra than does the Bulter-Reeds-Dawsons (BRD) method [193].

The spectra of a given sample can change with variations in the instrument itself and in the external environment (e.g., pH or temperature). It is well known that spectral variation can change key spectral features and ultimately reduce the identification accuracy of a model. Similarly, various spectroscopic devices can exhibit instrumental drift. Calibration transfer is a standardization technology that employs a multivariate calibration method to reduce the instrument- or environment-related spectral differences for the same or different spectroscopy instruments. Hou et al. proposed to use a 1D U-Net to replace the transformation matrix to achieve calibration transfer [188]. It was used to transfer a slave instrument spectral signal to a master instrument spectral signal. The results show that the calibration capability of 1D U-Net is superior to the Direct Standardization (DS) algorithm [194].

Shimming is an important method to reduce static magnetic field inhomogeneity, but the shimming process can be very tedious and complex, requiring operators to have a lot of experience. To improve efficiency, Becker et al. used DL to guide the shimming process [195]. Each input was composed of four groups of values of shim currents and their corresponding spectrum. The training process aimed to teach the DL network to transform these inputs into outputs that were the values of shim currents corresponding to a good spectrum. The results showed that DL can be used for fast first-order shimming in a low-field NMR device. Subsequently, Becker et al. [196] proposed an AI-driven shimming and traditional signal-based shimming method based on a LSTM network, and the speed and performance were significantly enhanced.

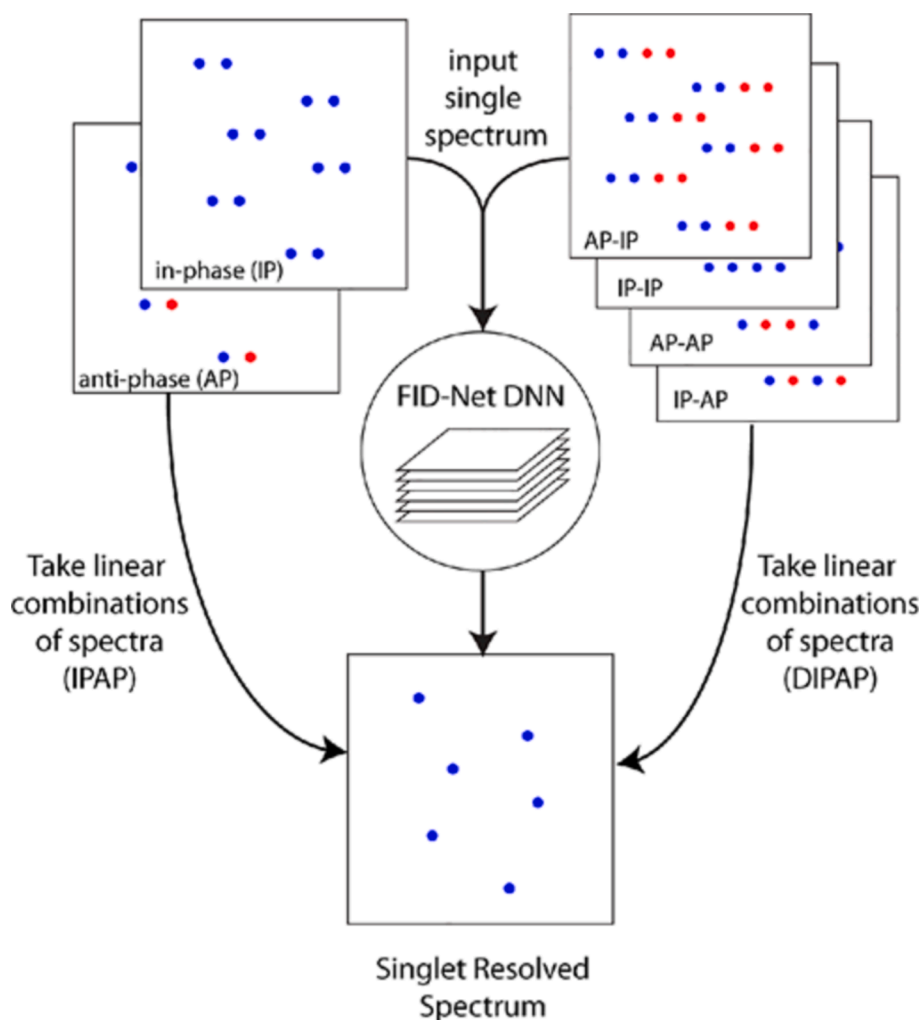


Fig. 20. Traditional methods, including IPAP and DIPAP, and the FID-Net-based method for virtual decoupling. Reproduced with permission from [201].

At present, as the authors said, shimming using DL is in its early stages, and its accuracy still has room for improvement.

DL has made some progress in low field NMR. It has the advantages of robustness and high data adaptability. However, reviewing the current research results, the application of DL in low-field NMR is still in its infancy, and most of the DL networks used have been relatively simple. The ultra-strong data processing capacity of DL has not been explored extensively.

4.8. Miscellaneous

In addition to the fields mentioned above, DL can also be used in other NMR applications. DL has become an effective alternative in NMR quantitative analysis, virtual decoupling, and instrument automation.

Yang et al. [197] designed a CNN-based network to process low-quality Zangger-Sterk pure shift spectral data, significantly improving spectral quality. In addition, Yang et al. designed a new loss function SM-CDMANE to increase the penalty for small errors in weak signal regions, enhancing the recognition ability for weak peaks. The trained network model can effectively remove artifacts up to a proportion of 14 % of a peak, distinguish weak peaks from noise with a signal-to-noise ratio as low as 3.6:1, and reduce line widths by more than a factor of three. Recently, to solve the problem of low sensitivity in pure shift NMR, Yang et al. [198] designed a CNN-based network named SE2CS-Net to obtain ^1H chemical shift information without strong coupling and

chunking artifacts by processing eight spin echo phase-changed spectra with different echo times. The sensitivity of the method matched that of conventional ^1H spectra, significantly surpassing current pure shift methods. These two works demonstrate the potential of deep learning in the development of pure shift technology. Kern et al. [199] applied DL in quantitative online NMR spectroscopy. In the past, reaction monitoring has relied on physical methods like Indirect Hard Modeling (IHM) [200], leading to time and cost-intensive processes. Kern et al. developed a new data augmentation approach to expand the dataset for neural network training by introducing systematic peak shape and position variations into the spectral model. As a result, the trained model was able to address moderately complex chemical synthesis challenges with four reactants. While DL offers results akin to IHM, it operates significantly faster, though it may miss changes beyond the training data space, like baseline distortions or unexpected components. In addition to quantifying reaction progress, DL can also be used to quantify NMR parameters like half-height peak width and chemical shift.

Recently, Karunanithy and Hansen [55] proposed FID-Net for reconstruction, providing a reference for DL in time-domain NMR processing. They used FID-Net to perform virtual decoupling on various spectra, including ^{13}C - ^{13}C correlation spectra of protein side chains, ^{13}CO - ^{15}N spectra, and $^{13}\text{C}\alpha$ - ^{13}CO protein correlation spectra [201,202]. In ^{13}C -detected biomolecular NMR methods, sensitivity and resolution losses due to one-bond ^{13}C - ^{13}C couplings are traditionally mitigated by virtual decoupling. This involves recording multiple spectra and taking

linear combinations, such as In-Phase Anti-Phase (IPAP) [203] and Double In-Phase Anti-Phase (DIPAP) [204]. The authors proposed a FID-Net based method to decouple directly detected spectra using only one spectrum (see Fig. 20). This method provided good decoupling performance in the case of one or two coupling constants. The performance of FID-Net was outstanding, but there were some problems. For example, the 1D spectra or slices need to have few peaks or couplings, and the processed spectra are not always clean enough, which hinders wider application and needs further improvement.

Soon afterward, Karunanithy et al. [205] utilized DL for analyzing Chemical Exchange Saturation Transfer (CEST) data. They focused on determining the amide proton chemical shifts of exchanging states using antiphase $^1\text{H}^{\text{N}}$ CEST profiles, which can be achieved in two steps: 1) transforming anti-phase to in-phase $^1\text{H}^{\text{N}}$ CEST profiles using a network built from the combination of a module similar to a block in the FID-Net framework and a modified LSTM module [54]; and 2) the estimation of chemical shifts, along with their uncertainties, using a network named DenseNet [66]. Compared with the traditionally used complex least-square fitting, the proposed DL method is free from the need for human experience and intuition, but it suffers from lower accuracy and only estimates chemical shifts, needing further improvement. Recently, they developed a new network named FID-Net-2 [206] to reconstruct high resolution ^1H - ^{13}C correlation maps of the aromatic side-chains in proteins and offer the uncertainty of reconstructed spectra (an indicator describing the reliability of spectra reconstructed by the network). This method has been experimentally validated on protein samples ranging in size from 7 to 40 kDa.

Cordova et al. [207] encoded a 2D dataset of Magic Angle Spinning (MAS) spectra recorded at various spinning rates into a sequence and employed an improved LSTM to infer isotropic proton spectra, obtaining high resolution spectra. Using relatively low MAS rates as the input can produce predictions with linewidths similar to fast MAS. If a wider MAS rate range and more MAS spectra are available, the model's performance is expected to improve [208].

With developments in NMR instrumentation, more and more experimental procedures have become automated. However, signal region determination for integration, one of the most routine steps in NMR spectroscopy, still typically needs to be performed manually. The 'int auto' command in the Bruker TopSpin software package can carry out automatic integration of the signal region, but the integration quality still relies on several manually selected parameters. Paruzzo et al. [209] developed a DL-based tool, called 'sigreg', for fully automatic signal region choice in 1D ^1H NMR spectra. The authors compared 'int auto' and 'sigreg' in different NMR spectra, and the results show that 'sigreg' had higher accuracy, with an F1 score of 94.8 %. The F1 score is defined as $2 \times (\text{precision} \times \text{recall}) / (\text{precision} + \text{recall})$, where precision is the ratio of true positive predictions to the total number of positive predictions (true positives plus false positives) and recall is the ratio of true positive predictions to the total actual positive instances (true positives plus false negatives). The difficulty in signal region determination is to identify signals in spectra containing many resonances, broad peaks, low SNR, high solvent signal intensity, or phase distortion. 'sigreg' is the first version, but already robust enough, and detecting signals in more complicated spectra needs to be addressed in later versions. Subsequently, Bruderer et al. [210] developed a DL-based tool, available as command 'apbk' in TopSpin, starting from version 4.1.3, for phase and baseline correction in 1D ^1H NMR spectra. Previous approaches, such as the frequently used 'apk/abs' combination, have struggled with high signal densities in this task. Both the DL-based 'sigreg' and 'apbk' routines exhibit robust performance, but they only work with 1D ^1H NMR spectra. In addition, Fischetti et al. [211] proposed a DL-based approach for detecting and classifying multiplets in ^1H NMR spectra of small molecules, offering a novel way to differentiate non-overlapping resonances in 1D spectra. However, it is not yet viable for spectra with overlapping resonances, like those of macromolecules.

Recently, Xiao et al. [212] designed a CNN-based network RH-UNet

for correcting distorted NMR spectra obtained in inhomogeneous magnetic fields. RH-UNet is effective for various solvent systems. However, it currently is not applicable to samples with large molecules. Methyl-TROSY NMR spectroscopy of partially perdeuterated proteins is effective for characterizing large molecules. However, the labelled protein synthesis is challenging and limits its application. To address this issue, Karunanithy et al. [213] utilized DL technology based on FID-Net for analysing spectra of protonated, uniformly ^{13}C -labelled proteins. This approach produces spectra of comparable quality to standard deuterated Methyl-TROSY spectra, potentially offering additional information. While this method is beneficial for studying large proteins and complexes without deuteration, it may result in decreased peak intensity.

DL can also assist in pulse sequence design. Manu et al. [214] optimized the TROSY-HSQC experiment using DL and evolutionary algorithms, resulting in significant improvements in sensitivity, acquisition speed, and other aspects. This offers a novel approach for accelerating biological macromolecular NMR experiments at high fields.

5. Application of deep learning in in vivo magnetic resonance spectroscopy

In vivo Magnetic Resonance Spectroscopy (MRS) is an important tool for non-invasively evaluating metabolite concentrations. This unique technique allows non-invasive exploration of the biochemical content, or metabolites, of specific organs without the need for surgical intervention or biopsy procedures. It tracks and measures the resonance frequencies and intensities of nuclear signals (proton signals are commonly used). Patterns of chemical shifts are unique to different metabolites, offering a 'biochemical fingerprint'. This fingerprint can help identify and differentiate between various physiological and pathological conditions.

Nowadays, in vivo MRS spectra can be measured in commercial MRI scanners, bringing convenience to the clinical setting. Typically, spectra are acquired using a short echo time to minimise decay from transverse relaxation (T_2) and J coupling modulation. The raw MRS data acquired are then be subjected to several steps of preprocessing, analysis and quantification to obtain reliable estimates of metabolite concentrations.

DL can be helpful in many aspects of MRS: denoising, frequency and phase correction, spectral quantification, and disease detection. This section will mainly introduce the applications of deep learning in the above four areas.

5.1. Magnetic resonance spectroscopy denoising by deep learning

5.1.1. Introduction to denoising of magnetic resonance spectra

Non-invasive acquisition of biochemical and metabolic information on tissues is a significant advantage of MRS techniques. However, the spectral SNR tends to be low, due to the low metabolite concentrations and restricted acquisition times in live subjects [215]. To improve the SNR and obtain high-quality spectra, it is necessary to reduce or even eliminate noise rather than simply increasing the Number of Signals Averaged (NSA) or increasing the magnetic field strength, both of which may not be practical in clinical applications.

Noise can be reduced by applying a smoothing filter in the spectral domain, such as a Gaussian filter, but filtering leads to partial information loss because it is challenging to fully distinguish between noise and signal. Further, filtering also causes broadening of the spectral lines, which reduces the spectral resolution [216]. Therefore, it is important to develop effective denoising techniques to improve the quality of MRS spectra. To address this issue, researchers have explored a variety of basic concepts and methods for denoising MRS signals. These methods encompass traditional techniques such as PCA [217], wavelet analysis [218] and low-rank matrix recovery [219], as well as the rapidly evolving and widely recognized DL methods.

Within the field of DL, a variety of cutting-edge denoising techniques have been applied to MRS signal processing [220]. Through the

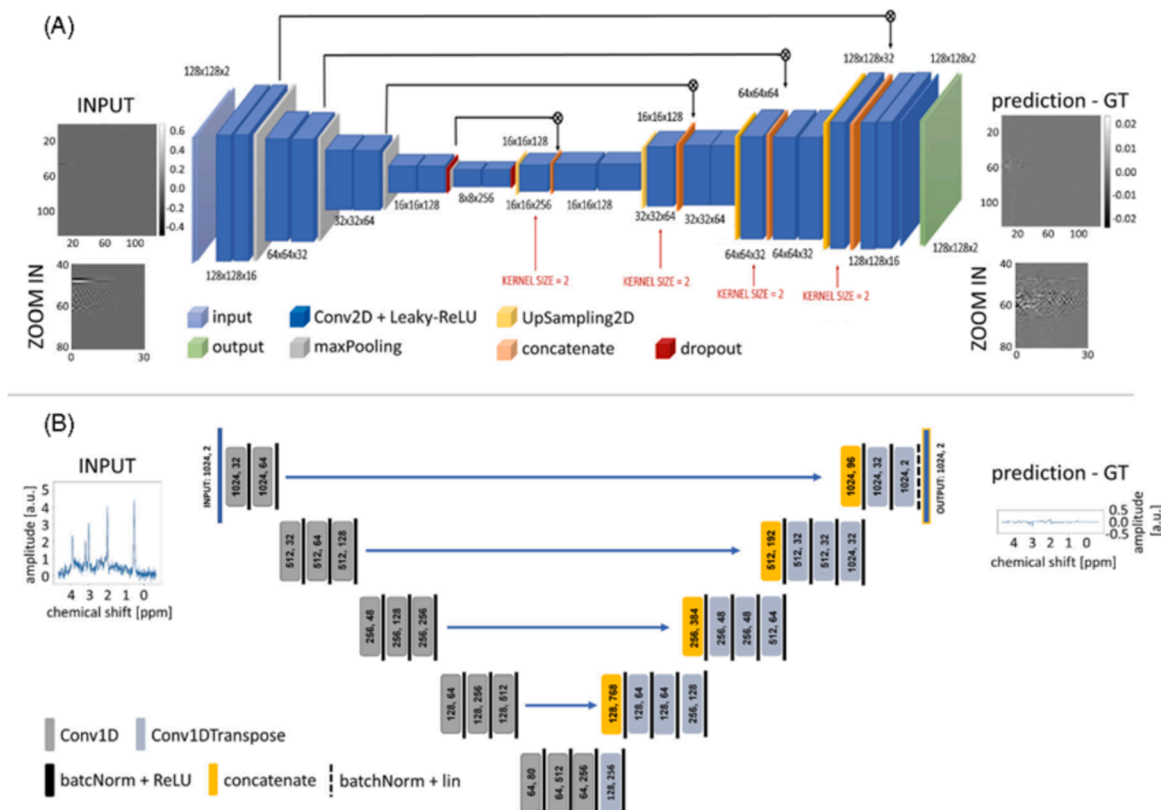


Fig. 21. Illustration of the DL denoising networks and corresponding sample data. (A) A sketch of the 2D-UNet, with relevant parameters indicated. (B) A sketch of the 1D-UNet, with relevant parameters indicated, with a sample input spectrum on the left, and the resulting residue spectrum on the right. Reproduced with permission from [220].

comprehensive exploration of these denoising methods, we are not only able to identify denoising methods suitable for specific noise environments, but also reasonably evaluate the efficacy of various methods according to specific application scenarios. This helps to extract metabolic information more accurately and promotes further applications of MRS in diagnosis and research.

5.1.2. Deep learning-based denoising for magnetic resonance spectroscopy

Lei et al. [221] were the first to introduce DL into MRS denoising, employing the Stack Auto-Encoder (SAE) network to remove noise from MRS data. The SAE can represent noisy data within a sparse feature vector field, diminishing noisy components. Results indicated significant SNR improvements after denoising with the SAE approach ($p < 0.05$) in low NSA spectra from both phantom and the parietal and temporal lobes of human brains. This approach preserves information on major metabolites (e.g., N-acetylaspartate, creatine, glutamine/glutamate, etc.) in *in vivo* spectra. SAE's sparse representation ability, not requiring prior knowledge, unlike compressed sensing, makes it ideal [222]. The potential contribution of SAE to clinical practice includes addressing principal challenges like low sensitivity and long acquisition time and preserving sufficient spectroscopic detail for the detection and quantification of target metabolites.

Typically, MRS scanning requires multiple samplings and averaging to achieve acceptable SNR, especially given the restricted magnetic field strength of clinical MRI scanners [223]. Chen et al. [224] proposed an LSTM-based [45] denoising model to reduce this problem. This network combines LSTM with a refusion mechanism and data validation, aiming to mine the correlations among time-domain data points and learn the mapping from low NSA (24 NSA) to high NSA (124 NSA) in the time domain. They combined different repetitions of a single voxel in *in vivo* MRS data to generate a more realistic training set. By selecting m repeated samples randomly out of M total repetitions, a significant

volume of training data were produced. This method demonstrated superior performance compared to the low-rank denoising method, reducing data acquisition time by up to 80 % and enabling fast single-voxel MRS.

Dziadosz et al. [220] used simulated ^1H MRS data to test different DL-based denoising methods. The inputs of MRS spectra come in two forms: time-frequency domain spectrograms, and 1D spectra. The corresponding networks were 2D-UNet and 1D-UNet, as shown in Fig. 21. The results showed that their DL-based denoising approach is beneficial solely for visualization purposes and does not contribute to quantitative assessments. So, denoising as a part of preprocessing needs further research.

5.2. Frequency and phase correction using deep learning

5.2.1. Introduction to frequency and phase correction in signal averaging

In MRS, a large number of transients are typically captured and averaged to enhance SNR. However, each transient may experience different frequency and/or phase shifts, due to hardware imperfections or physiological processes [225]. Averaging transients without Frequency-and-Phase Correction (FPC) can cause line-broadening and affect the signal line shape of the averaged MRS signal. Hence, FPC is often performed for each transient before averaging, especially when using spectral-editing MRS methods [226], to avoid subtraction artifacts.

Traditionally, FPC is done by adjusting two parameters – frequency and phase. FPC methods can be broadly classified into absolute and relative methods. The former corrects each transient individually, while the latter aligns transients to a reference signal. Popular FPC approaches include using the water peak as a reference [227] or fitting a metabolite peak to a model to estimate the shift(s) [228], spectral registration [229], and frequency-domain correlation [230]. Considering that FPC is

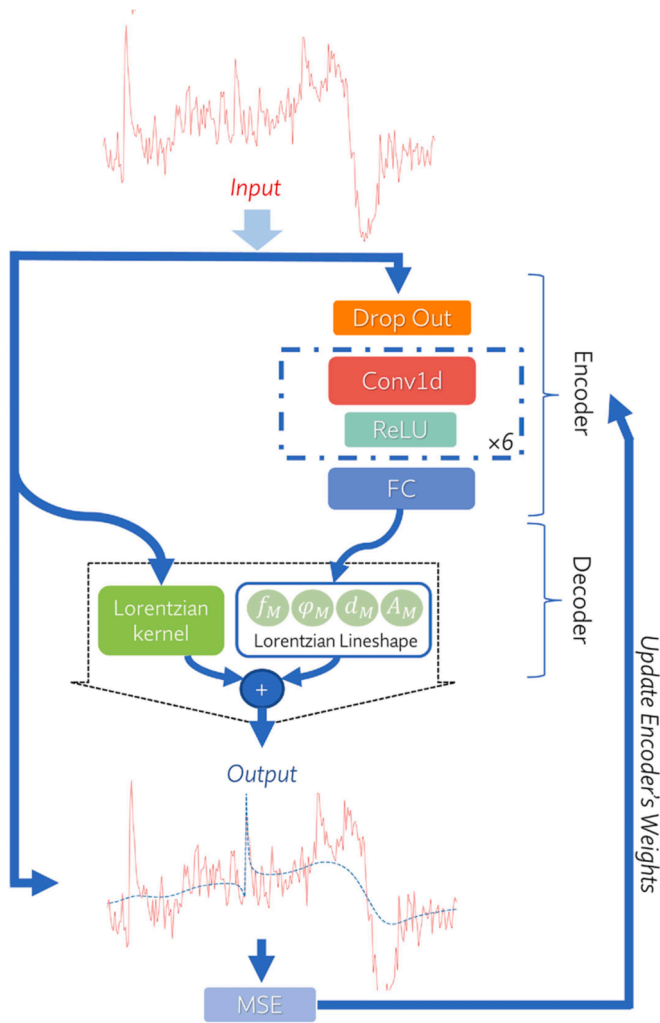


Fig. 22. Illustration of the network for the dCrR method. Abbreviations: A_M , amplitude; d_M , damping factor; f_M , resonance frequency; φ_M , zero-order phase of the Lorentzian lineshape. Reprinted with permission from [235].

achieved by adjusting frequency and phase, it is reasonable to anticipate that the variability across a set of signals collected for improving SNR should have a low-dimensional representation. Non-linear dimensionality reduction methods, such as manifold learning, can potentially exploit this, assuming that the high-dimensional data vectors are embedded in low-dimensional manifolds. This could open up new avenues like deep learning for improving FPC and thus MRS data quality.

5.2.2. FPC using deep learning

Tapper et al. [231] first proposed a fully connected neural network for FPC of Gamma-AminoButyric Acid (GABA)-edited MRS data. One year later, Ma et al. [232] proposed a CNN-based approach to achieve FPC of GABA-edited MRS signals. Recently, they [233] have also investigated a CNN framework for DL-based MRS spectral registration, with both supervised and unsupervised learning. The results show that the CNN approach has good performance in both simulations and in vivo datasets.

Bugler et al. [234] investigated the significance of complex-valued inputs and complex-valued convolutions compared to real-valued inputs and real-valued convolutions, in CNNs using both simulated and in vivo data. The results showed that in their models, using complex-valued inputs with complex-valued convolutions was most effective for FPC.

To estimate frequency and phase and achieve FPC, Shamaei et al.

[235] trained two Deep Auto-Encoder (DAE)-based networks, combined with Creatine Referencing (CrR) [228,236] and Spectral Registration (SR) [229], respectively. These two networks, Deep learning-based Creatine Referencing (dCrR) and Deep learning-based Spectral Registration (dSR), share similar structures and harness a parametric analytical method, embedding it within the DAE to estimate frequency and phase shifts in the transient. The networks were trained and validated on a simulated dataset with known ground truths, and tested on both phantoms and in vivo GABA-edited MRS data from the Big GABA repository [237]. Fig. 22 shows the framework of dCrR (dSR shares a similar network structure, except for a slight difference in the decoder). The input and output of dCrR and dSR are both complex signals in the time domain (the input and output are shown in the frequency domain for ease of understanding).

The difference between dCrR and dSR is mainly in the decoder part. The mathematical expression for the decoder in dCrR can be written as:

$$g(x; A_M, d_M, f_M, \varphi_M, t) = A_M e^{-d_M t} e^{-i(2\pi f_M t)} + x(t) e^{-d_L t}$$

where A_M , d_M , f_M , and φ_M , are the amplitude, the damping factor, the resonance frequency, and the zero-order phase of the selected Lorentzian lineshape, respectively, and d_L is the linewidth of the Lorentzian kernel.

For dSR, the mathematical expression of the decoder is given by:

$$g(f_M, \varphi_M, t) = R(t) e^{-i(2\pi f_M t + \varphi_M)}$$

where f_M and φ_M are the frequency and the phase shifts of the input signal with respect to the reference scan $R(t)$. In the dSR process, the encoder is designed to predict two parameters, namely relative phase and frequency shifts, simplifying the estimation from the traditional four parameters associated with line shape characteristics. Compared to traditional FPC methods like spectral registration, frequency-domain correlation, and creatine referencing, dCrR and dSR show similar or better performance and can offer efficient FPC of large MRS datasets in a shorter time.

5.3. Magnetic resonance spectroscopy quantification using deep learning

5.3.1. Introduction to magnetic resonance spectroscopy quantification

MRS has captivated biomedical researchers for several decades due to its ability to non-invasively measure tissue metabolites, which can be crucial for clinical diagnostics [238]. Such metabolic information may reveal pathway changes before anatomical alterations become evident, thus aiding in the early diagnosis of diseases [226,238]. However, the accurate quantification of metabolite concentrations is fraught with challenges, which limit the application of MRS in present clinical practice. Current MRS quantification techniques revolve around model fitting of the measured signal in either the time domain, the frequency domain, or both [239]. Although choice of domain for model fitting should not affect the outcome, in reality, discrepancies may arise. A common approach is to use a version of the Non-Linear Least Squares (NLLS) algorithm to estimate metabolite parameters using a prior basis set of metabolites.

LCModel [240] is often taken as the gold standard in spectral quantification. It employs NLLS to break down spectra in the frequency domain, and can directly simulate macromolecules. On the other hand, TARQUIN [241], VeSPA [242], and jMRUI [243] can simulate their basis sets for quantification, providing a broader range of tools for analysis. However, despite all the methods mentioned here, reliable and accurate quantification of MRS still remains a challenge.

5.3.2. MRS quantification: A deep learning approach

MRS quantification presents a considerable challenge, owing to the low SNR, chemical shift variability, and overlapping signal components. A study by Hatami et al. [244] demonstrates how DL, specifically 1D CNNs, can be utilized for MRS signal quantification.

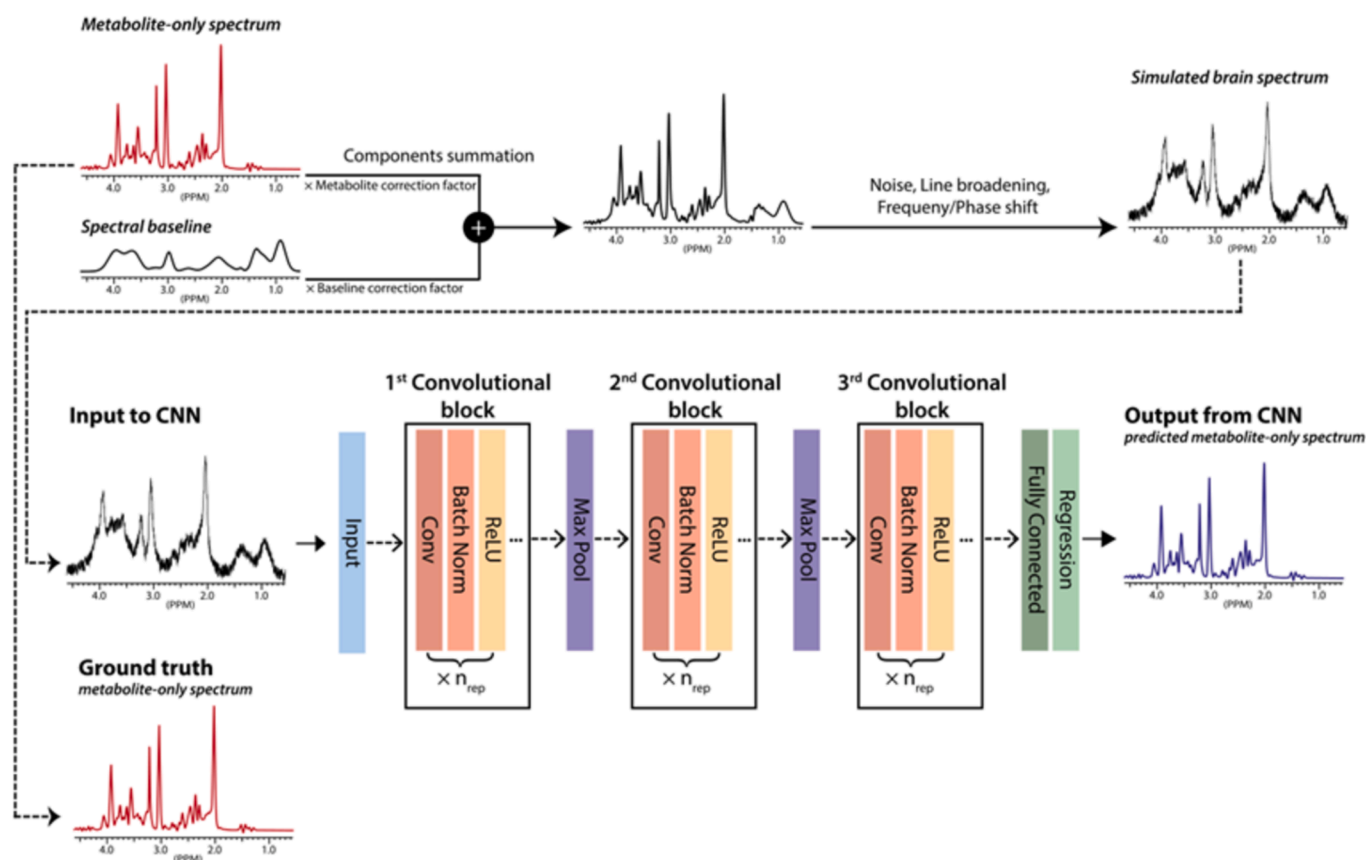


Fig. 23. Illustration for the simulation of spectra and the training of the CNN. Conv, convolution; Batch Norm, batch normalization; MaxPool, max pooling; ReLU, rectified linear unit; nrep = 4. Reproduced with permission from [245].

Lee et al. [245] trained CNNs on a dataset of approximately 40,000 spectra. They mined clean (intact) metabolite spectra and then quantified the metabolite-only spectra by solving simple inverse problems, as shown in Fig. 23. The quantification results show very good performance even at low SNR. In another study, Chen et al. [246] proposed QNet, which is the first Linear Least Squares (LLS) quantification aided by DL. QNet first applies multi-layer convolutions to predict the Imperfection Factors (IF) and the MacroMolecular (MM) background signal, separately, to simplify the problem of metabolite quantification.

It is important to note the differences in quantification between the time and frequency domains. For example, eliminating the first few distorted data points in the time domain does not significantly affect time-domain analysis, but omitting data points can induce complex modulations across the entire spectrum in the frequency domain [226]. The use of time–frequency analysis has been advocated [247] to address this limitation. Despite its use across several fields over the years, it remains challenging to find a precise tool for this type of analysis. DL shows great promise in this regard, as exemplified by Thomas et al., where time–frequency images of a speech signal were used as input to a CNN for classification [248]. Similarly, Kyathanahally et al. [249] trained a CNN with time–frequency data to detect and eliminate ghosting artifacts in clinical magnetic resonance spectra of the human brain.

To reduce the computational cost of CNN, Shamaei et al. [250] proposed a Wavelet Scattering Convolutional Network (WSCN) for metabolite quantification. It was trained and tested only on simulated data and needs further validation on in vivo data. The studies above use supervised learning. Shamaei et al. [251] proposed an LCM-based (linear combination model) self-supervised DL method for the quantification of MRS signals. This method shows comparable performance

with conventional approaches but in a shorter time. These developments underscore the potential utility of DL in the advanced analysis and processing of MRS signals.

The studies mentioned above primarily focus on quantifying spectra acquired using the Point-Resolved Spectroscopy (PRESS) sequence. The PRESS sequence, while efficient, often hinders the identification and quantification of hard-to-detect metabolites such as GABA. Chandler et al. proposed a CNN named MRSNet for metabolite quantification in edited MRS [252]. They trained the network with simulated data, and validated it using calibrated phantom spectra. The network needs further validation to ensure accuracy and reliability.

Performance validation of such DL algorithms on in vivo spectra is mainly based for comparisons with LCModel quantification results. While LCModel is a widely accepted method, this highlights the need for independent validation to fully assess the accuracy and reliability of DL models. Huang et al. [253] compared CNN-based methods and LCModel for the quantification of in vivo MRS data. The results showed that the CNN-based method has the potential to serve as a quantification tool for in vivo MRS spectra. However, the CNN-based method does not yield reliable quantification results for metabolites with complex spectra. Rizzo et al. [254] explored the reliability of the quantification of MR spectra by deep learning in an idealized setting. They found the model fitting method mostly performed better than the faster DL approach in their study.

For 2D MRS, Iqbal et al. [255] used DL not only to quantify but also to accelerate simulated localized correlated spectroscopy (L-COSY). This approach showed accurate reconstruction and quantification for most SNR levels. However, it needs further research to validate this method for in vitro and in vivo situations. For *J*-Point Resolved Spectroscopy (JPRESS), in which a series of different TE spectra are acquired, Zhang

et al. [256] proposed an end-to-end DL network to quantify absolute metabolite concentrations without spectral fitting. In addition, to enhance the convenience of DL in MRS quantification, Chen et al. [257] built a cloud-based online platform called CloudBrain-MRS to integrate preprocessing, quantification, and analysis.

Recent studies show promising results in deploying DL for interpreting complex MRS data. However, these methods are limited to specified sequences, reliable references are still unavailable, and the robustness and reliability of these models need to be evaluated. Extending the use of DL beyond the conventional PRESS sequence, introducing novel DL architectures, employing independent validation strategies, and enhancing model reliability are some of the exciting future prospects in MRS quantification using DL.

5.4. Disease detection by deep learning using magnetic resonance spectroscopy

5.4.1. Introduction to disease detection using magnetic resonance spectroscopy

MRS has the potential to identify metabolic abnormalities prior to the appearance of structural lesions. Its adaptability enables the examination of diverse metabolic functions in various tissues. While mainly employed for brain analysis, MRS is also effective in detecting, localizing, staging, assessing aggressiveness, and evaluating the treatment response of, cancers in the breast, prostate, liver, and other organs.

Through the analysis of brain MRS data, the presence of disease or lesions in the brain can be detected, and screening can be performed for detecting early-stage diseases such as cancer and Alzheimer's disease (AD). AD is the most common form of dementia, and its diagnosis is particularly challenging in its early stages, such as Mild Cognitive Impairment (MCI) and Mild Alzheimer's Disease (MAD) [258], where structural brain atrophy is difficult to detect. Neuroimaging techniques, including anatomical MRI and ^1H -MRS, offer potential solutions to this challenge. Although anatomical MRI is a common tool for assessing structural brain damage, it may not be sensitive enough for early detection of AD. In contrast, ^1H -MRS has the potential to reveal key brain metabolic information, and can non-invasively detect early biological changes in the brain [259]. Metabolic changes detected in specific brain regions, such as the posterior cingulate cortex, may serve as potential early biomarkers for AD, helping to identify brain abnormalities before structural damage and clinical symptoms appear. For example, by analyzing cases of metastatic and glioblastoma brain

tumors from the INTERPRET dataset [260], integrating neuroimaging techniques like ^1H -MRS has huge potential. However, the application of MRS in disease detection also faces many challenges, particularly noise in the data, which may lead to spectra from healthy brain tissue being incorrectly labeled as 'tumor'.

5.4.2. Disease detection by deep learning

To improve the accuracy of tumor detection using MRS data, Lu et al. [261] introduced a DNN-based framework that competes effectively with human experts in distinguishing between tumor and non-tumor tissues. To overcome the challenge of misclassification, they proposed a two-step data distillation and augmentation procedure. First, the data are purified using a neural network; second, the data are augmented to enhance the training set for a second neural network, which performs the final classification. This method has demonstrated broad potential, and could be applied across various research fields. However, a limitation of the framework is that it processes only a single spectrum per patient. Future consideration should be given to patient-wise training, which would utilize multiple spectra from the same individual. Despite this, Lu et al.'s [261] framework made a significant stride toward enhancing clinical practice and increasing the accuracy of brain tumor diagnoses.

In the pursuit of enhancing early AD detection, Ahmed et al. [262] proposed an end-to-end deep learning network for early AD and Normal Control (NC) subject classification using ^1H -MRS raw data. They introduced a novel multi-class classification framework leveraging ^1H -MRS data [263]. Central to this approach was an attention mechanism that strategically directs the model to metabolites with high diagnostic relevance, thereby improving class discrimination. The framework utilizes synthetic data to surmount the challenges posed by data scarcity, aiding in the effective learning of the feature space. The results showcase the method's superiority, outperforming the current state-of-the-art method (DeepMRS [262]) in various extensive metrics. Particularly, this method outperforms imaging biomarker-based approaches, underscoring the potential of metabolite feature learning in early AD detection. The integration of ^1H -MRS data with 1D-CNN heralds a promising future for establishing a new virtual biopsy technique. Fig. 24 illustrates this multi-class classification framework, signifying a milestone in AD research.

In the context of pseudo brain tumors, stacked LSTM and Bidirectional LSTM (Bi-LSTM) have been employed to enhance diagnostic accuracy. The performance of these trained models has been compared

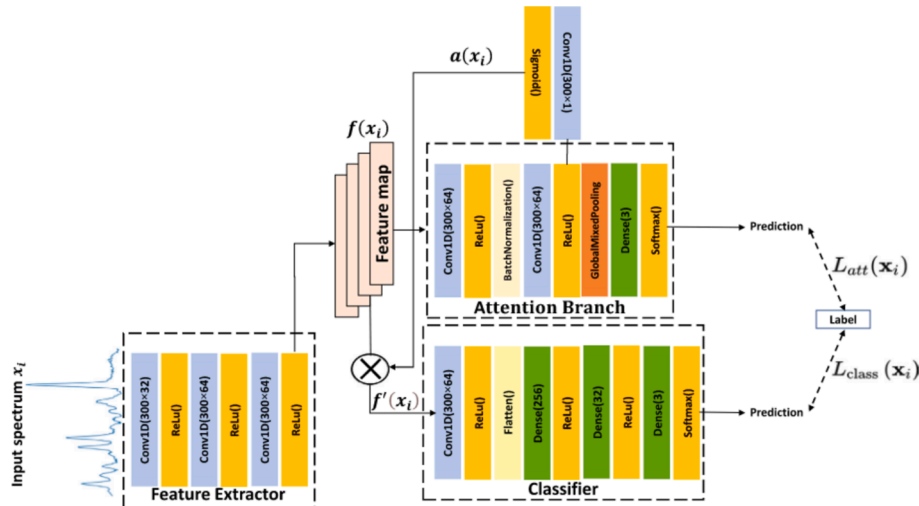


Fig. 24. Illustration of a self-guided attention-intensive deep network. The attention block employs an attention framework that instructs the model to concentrate on significant metabolites and ignore non-pertinent ones, for class differentiation in a governed manner. The attention block is crafted to take as input feature maps $f(x_i)$ to construct an attention map $a(x_i)$ which is subsequently employed to concentrate on distinguishing segments of signal in a holistic learning procedure. Reproduced with permission from [263].

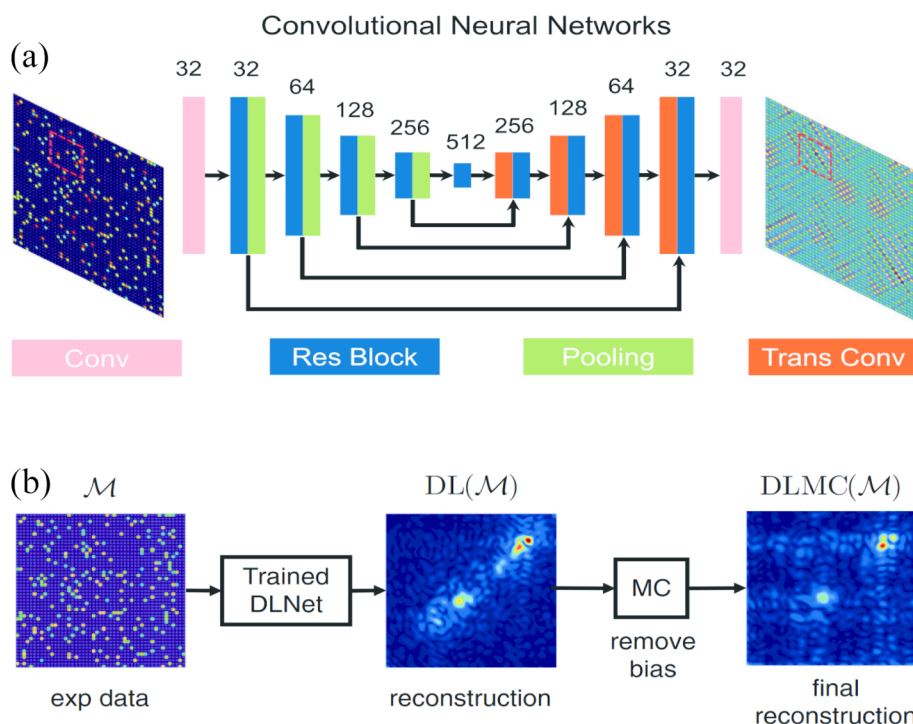


Fig. 25. (a) Schematic diagram of CNN reconstruction, and (b) schematic diagram of the overall DLMC. Reproduced with permission from [281].

with the that of 1D-CNN, with the LSTM variants showing superior results across binary classifications of the spectra from normal brain tissues, glioblastoma, metastatic, and diffuse astrocytoma brain tumors. High accuracy rates were achieved, ranging from 85.6 % to 99.2 %, despite the arguably marginal differences in results between the LSTM model and its bidirectional variant. The trained models' accuracy reflects their efficacy across multiple sets, reinforcing their reliability. The scarcity of MRS signal datasets, particularly for pseudo brain tumors, necessitated the use of data augmentation techniques. This involved the generation of unique yet minimally differentiated MRS data through a randomized procedure, which ultimately enhanced the detection accuracy of pseudo brain tumors.

Deep learning has also achieved good results in volume selection [264], artifact removal [249], uncertainty measurement [265,266], reconstruction [267,268], artificial data generation [269], and quality assessment of MRS spectra [215,270]. Future work may consider alternative data augmentation methods, fine-tuning network parameters, or integrating CNN with LSTM networks for enhanced classification capabilities. Critical to the advancement of this research area is the creation of larger MRS datasets, which can facilitate a more comprehensive evaluation of a proposed method's accuracy.

6. Deep learning in nano NMR

Traditional NMR relies on large-scale molecular ensembles to obtain adequate SNR, which sacrifices specific information about individual molecules. Due to the high sensitivity of atomic-scale Nitrogen-Vacancy (NV) centers, single-spin-sensitive NMR, single-molecule MR, and microscopic two-dimensional NMR can be achieved with NV centers [271–278]. DL also has excellent performance in this field of nanoNMR [279–281].

Like traditional NMR, nanoNMR suffers from inherent noise, making it difficult to achieve optimal spectral identification, especially for weak peaks. Aharon et al. [280] used a DL model, denoted M_{DL} , to mitigate the effect of the inherently strong noise in nano-NMR settings. Experimental results showed that whether or not the noise model was precisely known

(it is complex even if known), the performance of M_{DL} is better than Bayesian methods.

In addition, nanoNMR has the same problem as traditional NMR that the collection of two-dimensional NMR spectra is very time-consuming. Therefore, it is desirable to non-uniformly sample the nanoNMR signal and then reconstruct it. Kong et al. [281] proposed a Deep-Learning-Matrix-Completion (DLMC) method based on the residual network, as shown in Fig. 25. Since DL requires a large amount of data for training, simulated data were employed. However, this way of network training brings with it the problem of domain shift. To solve this problem, Kong et al. used the traditional Matrix Completion (MC) method to post-process the output of DL to maintain a low-rank nature, so the domain shift problem in simulated data are alleviated to a certain extent. While this method performs well in nanoNMR reconstruction, it is anticipated that incorporating more information, such as data consistency (the sampled point data should not change significantly during the training process) could lead to better results.

7. Conclusion and outlook

This review comprehensively summarizes recent applications of DL in the acquisition, processing and analysis of traditional NMR, MRS, and nanoNMR spectra. In traditional NMR, DL has enabled tasks such as NUS reconstruction, denoising, peak picking, chemical parameter inference, compound identification, etc. In the field of MRS, DL has been applied to in vivo MRS for denoising, frequency and phase correction, quantification, and disease detection. Additionally, in nanoNMR, DL has successfully achieved denoising and reconstruction. Compared to traditional algorithms, DL methods significantly improve the efficiency of spectral processing. These methods can automatically perform many complex tasks, reducing manual intervention, enhancing automation, and demonstrating robust performance. Several critical issues remain unresolved. For instance, it is difficult to evaluate the quality of reconstructed spectra without fully sampled spectra to compare them with. Additionally, in noise suppression, the intensities of denoised peaks cannot be precisely restored. Addressing these challenges is crucial for

future advancements.

To achieve optimal performance in deep learning applications in the NMR field, it is crucial to focus on four key elements: training dataset, network architectures, hyperparameters, and loss functions.

Training Dataset: Training deep learning models requires a large amount of data. Currently, getting a lot of real experimental data are often impractical, so most methods use simulated data for training. However, differences between simulated and real data do exist, often causing suboptimal performance of DL on real data. Therefore, using few-shot learning techniques [282] to train models on real data may produce better results.

Network Architecture: The DL models commonly used in the NMR field are primarily based on classical network architectures such as U-net, ResNet, WaveNet, etc. Although some studies have explored more advanced architectures like Transformers, further developments in advanced networks are still needed. As the field of DL continues to advance rapidly, it is good to stay informed about new developments and assess their potentials on NMR spectroscopy.

Hyperparameters: Hyperparameters have a huge influence on the performance of a model. Effective tuning of hyperparameters such as learning rates is crucial for achieving optimal results. Future studies exploring automated hyperparameter optimization techniques to facilitate model training are needed.

Loss Functions: The choice of loss function is fundamental to the learning process. Different tasks may have different optimal loss functions. Future research focusing on developing specialized loss functions for the unique demands of NMR applications is needed.

In conclusion, DL shows great potential in the fields of traditional NMR, MRS, and nanoNMR, but it also faces challenges. Through continuous technological innovation, DL is expected to further improve the efficiency and accuracy of its application in NMR, providing stronger support for scientific research and practical applications.

CRediT authorship contribution statement

Yao Luo: Writing – review & editing, Writing – original draft, Formal analysis. **Xiaoxu Zheng:** Writing – review & editing, Writing – original draft, Formal analysis. **Mengjie Qiu:** Writing – review & editing. **Yaoping Gou:** Writing – original draft. **Zhengxian Yang:** Writing – original draft. **Xiaobo Qu:** Resources. **Zhong Chen:** Funding acquisition. **Yanqin Lin:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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